When Money Crowds Out Morals: A Study of Demand-Side Management Participation

Yuan Zhao* and Dingwei Guo†‡

Version: July 11, 2026

Abstract

This study investigates how extrinsic incentives interact with intrinsic prosocial motivation in shaping household participation in Direct Load Control (DLC) programs. Building on a theoretical framework of motivation crowding-out, we combine two complementary experiments to examine whether and how extrinsic incentives affect prosocial participation. The first experiment uses a discrete choice experiment (DCE) to elicit respondents' preferences for key DLC attributes and provides suggestive evidence of a U-shaped relationship between compensation and DLC acceptance. Motivated by these findings and to address potential identification concerns inherent in the DCE design, the second experiment uses a randomized design that varies monetary incentives. We find that prosocial motivation encourages DLC participation, while small monetary incentives can erode this motivation and reduce participation rates. However, sufficiently large monetary incentives can reverse this effect by dominating the decision-making process. These results underscore the importance of carefully balancing intrinsic motivation and extrinsic incentives in the design of demand-side management (DSM) programs.

Keywords: Demand response, motivation crowding-out, prosocial values, discrete choice experiment

JEL Codes: D91, C93, C25, Q41

*Jinhe Center for Economic Research, Xi'an Jiaotong University. Email: zhaoyuan@xjtu.edu.cn

†Corresponding author. Department of Economics, University of North Carolina at Chapel Hill. Email: dwguo123@outlook.com

‡This research is based on the author's undergraduate thesis in 2022. We appreciate the funding provided by the Soft Science Foundation in Shaanxi Province, China.

1 Introduction

As electricity demand rises and concerns about climate change and energy sustainability intensify, demand-side management (DSM) has become an important tool for reducing peak demand by allowing utilities to directly control household loads during high-use periods. Yet the global expansion of direct load control (DLC) and other DSM programs remains far below the level needed for decarbonization. In the IEA's Net Zero Scenario, DSM and storage together are projected to provide only about one-quarter of global flexibility by 2030 and half by 2050, and survey evidence shows that participation in DLC programs often remains below 5% (Faruqui and Sergici, 2010; Newsham and Bowker, 2010; Stenner et al., 2017).

Barriers to residential participation in DLC programs span economic (Meyer et al., 2021; Panda et al., 2022; Scharnhorst et al., 2024), institutional (D'Ettorre et al., 2022; Mohanty et al., 2022; Bakare et al., 2023), and behavioral dimensions (Perlaviciute and Steg, 2014; Frederiks et al., 2015; Uddin et al., 2018; Williams et al., 2023). To address these barriers, most DLC programs rely on financial incentives, such as monetary rewards. Using contingent valuation methods (Oerlemans et al., 2016; Kaczmarski et al., 2022; Zhao et al., 2022) and discrete choice experiments (Morrissey et al., 2018; Zha et al., 2020; Brennan and van Rensburg, 2023; Bender et al., 2024), prior studies estimate households' willingness to pay (WTP) or willingness to accept (WTA) for DLC under alternative pricing schemes. Beyond economic incentives, survey-based studies in economics and energy document that prosocial and pro-environmental attitudes can also encourage DSM participation, such as positive attitudes toward environmental protection and energy saving (Familia and Horne, 2022; Sloot et al., 2022; Skoczkowski et al., 2024; Fabianek et al., 2025).

Most of the literature implicitly treats prosocial motivation and monetary incentives as complementary. For example, engineering simulations that incorporate social factors into agent-based models show that neighborhood interactions can increase DSM adoption and reduce the monetary incentives required by up to 30% (Blanke et al., 2017; Sridhar et al., 2024; Golmaryami et al., 2024). Behavioral economics, however, highlights a potential conflict. Introducing any extrinsic monetary incentives may reduce participation driven by prosocial motives and increase participation driven by monetary motives. When rewards are modest, however, the latter increase may be insufficient to offset the former decline, leading to lower overall participation (Vohs et al., 2006).

This crowding-out effect has been documented in diverse domains, including acceptance of nuclear waste repositories (Frey et al., 1996), blood donation (Mellström and Johannesson, 2008), market transactions (Bartling et al., 2015), environmental donations (d'Adda, 2011; Bartling et al., 2015), electricity conservation (Steinhorst et al., 2015; Goff et al., 2017), and laboratory experiments (Gneezy and Rustichini, 2000; d'Adda et al., 2020). Theoretical models of conformity and prosocial behavior under incomplete information further provide microfoundations for how extrinsic incentives can interfere with intrinsic motivation (Bernheim, 1994; Bénabou and Tirole, 2006; Fischer and Huddart, 2008; Ali and Bénabou, 2020; Battigalli and Dufwenberg, 2022). Despite this extensive literature, evidence on crowding-out mechanisms in residential DLC remains scarce in both engineering and economics, leaving an important gap that this study addresses.

In this paper, we examine how monetary incentives and prosocial motivation jointly shape household participation in DLC programs. Building on the theoretical framework of Bénabou and Tirole (2006), we argue that accepting monetary compensation for DLC participation may signal greediness to their neighbors, generating reputational losses that discourage participation when compensation is positive but low. As compensation rises, the monetary benefit can offset the reputational cost and increase participation. This interaction gives rise to a U-shaped relationship between participation and compensation, in which participation initially declines when compensation increases from zero to a low level, but rises once compensation becomes sufficiently large.

To test these arguments empirically, we study a DLC program targeting residential air conditioner (AC) use through an online survey and a randomized experiment conducted in Beijing and Shanghai, China. The first survey with 775 respondents uses a discrete choice experiment (DCE) to elicit respondent preferences for DLC attributes, including temperature restrictions, episode duration, data-storage policies, and monetary compensation. Unlike standard DCEs that ask respondents to choose only their most preferred alternative, our design asks them to rank three alternatives consisting of two DLC alternatives and the status quo (opting out of DLC) in each choice card. This ranking format allows us to elicit respondents' respective preferences for each DLC alternative relative to the status quo.

Using the random-parameter logit (RPL) model, we estimate the average WTA for each attribute, which characterizes how respondents value different features of DLC programs and informs the program pricing in the Chinese context. Building on these estimates, we then use the ranking format to examine how monetary incentives interact with prosocial motivation. We find suggestive evidence of crowding-out, as shown by an empirical U-shaped relationship between compensation and DLC acceptance relative to the status quo.

Since the DCE includes a rich set of hypothetical DLC alternatives, identifying crowding-out effects from the DCE can help reduce the concern that the estimated effect is driven by preferences for a particular DLC alternative shown to respondents. However, the challenge of identification based on the DCE is that we cannot include a zero-compensation level in any choice cards, which prevents identifying the baseline acceptance rate in the absence of monetary incentives. Choice modelling requires a monetary compensation attribute to recover WTA and translate attribute trade-offs into preference measures, but introducing this attribute also makes the financial dimension salient in every choice task (Hanley et al., 2001). Once respondents observe the monetary compensation attribute in the DCE, they may begin evaluating hypothetical DLC alternatives through a more transactional frame, which can weaken intrinsic prosocial motivation even when a zero-compensation option is displayed.

To address this concern, we implement another randomized incentive experiment that varies extrinsic incentives for a second group of respondents. We sampled 647 new respondents in the two cities to avoid the contamination from the DCE. Respondents are shown one representative DLC alternative and randomly assigned to one of five incentives following the attribute setting in the DCE: voluntary participation with no compensation, 50 CNY compensation, 100 CNY compensation, 200 CNY compensation, 400 CNY compensation. We find that participation is significantly lower in the 50 CNY, 100 CNY and 200 CNY groups than in the voluntary no-compensation group, while it is higher in the 400 CNY group. This result provides complementary and robust evidence of crowding-out in DLC programs.

In the final part of the paper, we use the randomized experiment results to structurally estimate the parameters of the Bénabou and Tirole (2006) model, which allows us to visualize the U-shaped relationship and conduct counterfactual policy simulations. The simulations show that reducing heterogeneity in self-interest mitigates crowding-out effects.

This study contributes to three strands of research. First, it contributes to the literature on barriers to and motivations for DSM participation. To the best of our knowledge, this is among the first papers to explicitly examine crowding-out effects in the DSM context. Beyond the prosocial and pro-environmental motivations, most DSM studies on non-monetary motivation focus on personal benefits (Gyamfi and Krumdieck, 2011; Shipman et al., 2013; Lynch et al., 2019; Roldán-Blay et al., 2019; Gamma et al., 2021; Lashmar et al., 2022; Yilmaz et al., 2022; Sridhar et al., 2023), innovation preferences (Bradley et al., 2016; Kowalska-Pyzalska, 2018; Hackbarth and Löbbe, 2020; Nilsson and Bartusch, 2024; Wen et al., 2025), and other behavioral factors (Hall et al., 2016; Anwar and O'Malley, 2021; Gleue et al., 2021; Skoczkowski et al., 2024), without considering their potential interactions with extrinsic incentives.

The two most closely related studies are Gamma et al. (2021) and Sloot et al. (2022). Gamma et al. (2021) study how responses to incentives and penalties vary by individual attitudes, while Sloot et al. (2022) examine how environmental framing affects acceptance among environmentally versus financially motivated individuals. However, both studies provide only limited discussion of crowding-out and do not directly examine its formation or mechanism in the DSM context.

Second, this study provides a baseline framework for DLC design and broader DSM strategies in China's residential electricity market. Although China's National Development and Reform Commission issued DSM guidelines in 2010, 2017, and 2023, academic research on residential DSM remains limited, and DSM programs in China are still at an early stage of promotion and implementation (Ming et al., 2013; Zhao et al., 2015; Zhou and Yang, 2015; Khanna et al., 2016; Zhang et al., 2017).

Finally, the findings contribute to the stated choice literature by highlighting a potential limitation of linear or monotonic specifications in stated choice models. Conventional choice-experiment models typically assume that higher payments monotonically increase utility, an assumption also common in the energy valuation literature (Daniel et al., 2018; Morrissey et al., 2018; Georgarakis et al., 2021; Kim et al., 2023; Bender et al., 2024). Our results challenge this assumption by showing that respondents may react discontinuously or non-monotonically to payment levels when monetary incentives alter the decision frame. These findings are consistent with recent discussions of nonlinear specifications in stated choice models (Dugstad et al., 2021; Ladenburg et al., 2022) and underscore the importance of incorporating behavioral insights into stated choice valuation.

The remainder of the paper proceeds as follows. Section 2 introduces the theoretical framework of crowding-out effects, drawing on the model of Bénabou and Tirole (2006). Section 3 describes the survey design and data collection. Section 4 outlines the empirical strategy. Section 5 presents the main findings. Section 6 concludes and discusses policy implications.

2 Theoretical Framework

We draw on the theoretical model of Bénabou and Tirole (2006) to illustrate how extrinsic incentives can crowd out intrinsic motivation and to motivate our survey design.

2.1 Model Set Up

Suppose household i chooses contribution x to a public program[1]. The first component of personal benefit comes from the household's intrinsic satisfaction from making this contribution and the extrinsic gain from obtaining the compensation, which can be written as:

$$(v_s + v_m m)x \tag{1}$$

where v_s is the intrinsic prosocial value derived from each unit of contribution x. It captures the household's sense of moral satisfaction from acting in a socially responsible manner and contributing to the community, such as its perceived social benefits of DLC participation. v_m captures the marginal utility from increased monetary payoff, and m denotes the amount of this compensation, such as monetary rewards.

The model assumes that both v_s and v_m vary across households and are private information that their neighbors cannot observe. However, the distributions of v_s and v_m are common knowledge. Each household's (v_s, v_m) is drawn from a population distribution.

Household i chooses x given m, and the resulting pair (x, m) is observed by neighbors and sends a signal about household i's private values. After observing this signal, neighbors form beliefs about i's v_s and v_m through $\mathbb{E}(v_s \mid x, m)$ and $\mathbb{E}(v_m \mid x, m)$. These beliefs then enter household i's utility through reputational concerns. Following Bénabou and Tirole (2006), this term is specified as:

$$\mu_s \mathbb{E}(v_s \mid x, m) - \mu_m \mathbb{E}(v_m \mid x, m) \tag{2}$$

[1] In our empirical setting, x indicates whether a household participates in the DLC program. For analytical simplicity, the theoretical model treats x as continuous.

where $\mu = (\mu_s, \mu_m) \geq 0$ captures the strength of reputational concerns. The coefficient on μ_m in equation (2) is negative because $\mathbb{E}(v_m \mid x, m)$ captures how greedy household i is perceived to be by neighbors, which is socially disliked. Following Bénabou and Tirole (2006), we assume that (v_s, v_m) follows a normal distribution $N(\bar{v}, \Omega)$.

Combining equations (1) and (2), the household's maximization problem is:

$$\max_x \; (v_s + v_m m)x + \mu_s \mathbb{E}(v_s|x, m) - \mu_m \mathbb{E}(v_m|x, m) - \frac{kx^2}{2}$$

where $kx^2/2$ is the cost of contributing.

2.2 Model Implications

For simplicity, we treat x as a continuous variable, which yields the optimal choice x^* as:

$$x^* = \frac{v_s + v_m m}{k} + \mu_s \rho(m) - \mu_m \chi(m) \tag{3}$$

where

$$\rho(m) = \frac{\sigma_s^2 + m\sigma_{sm}}{\sigma_s^2 + 2m\sigma_{sm} + m^2\sigma_m^2}$$

$$\chi(m) = \frac{1 - \rho(m)}{m}$$

and σ^2's are the variance and covariance in the covariance matrix Ω for (v_s, v_m).

Therefore, the average contribution in the population is:

$$\bar{x}^* = \frac{\bar{v}_s + \bar{v}_m m}{k} + \mu_s \rho(m) - \mu_m \chi(m) \tag{4}$$

where we assume that μ_s and μ_m are the same across households.

Equation (4) implies the following condition and proposition for crowding-out:

Proposition 1 *Crowding-out effects (Bénabou and Tirole (2006) Proposition 2):*

Let $\sigma_{sm} = 0$ and define $\theta = \sigma_m/\sigma_s$. Compensation m crowds out intrinsic motivation (i.e.,

$\partial\bar{x}^*/\partial m < 0$) *when* m *satisfies:*

$$\frac{\bar{v}_m}{k} < \mu_s \frac{2m\theta^2}{(1+m^2\theta^2)^2} + \mu_m \frac{\theta^2(1-m^2\theta^2)}{(1+m^2\theta^2)^2}.$$

Let $[m_1, m_2]$ *denote the set of compensation levels satisfying this inequality, over which* $\partial\bar{x}^*/\partial m < 0$*. This crowding-out interval with* $m_1 < 0 < m_2$ *does not always exist, and it arises only for certain values of the parameters* μ*,* $\bar{v}$*,* Ω*, and* k*.*

Proposition 1 characterizes the relationship between $\bar{x}^*$ and m when the crowding-out interval $[m_1, m_2]$ with $m_1 < 0 < m_2$ exists. When $m = 0$, positive contribution is driven entirely by intrinsic motivation, v_s. When a positive extrinsic incentive m is introduced, however, contribution can generate an additional reputational cost by lowering $\mu_s\mathbb{E}(v_s \mid x, m) - \mu_m\mathbb{E}(v_m \mid x, m)$. When m is positive but small, the monetary benefit, $v_m m$, is not large enough to offset this reputational cost, leading to a decline in contribution x. As m increases further, the monetary benefits eventually dominates and raises contribution. This U-shaped relationship between contribution x and extrinsic incentive m illustrates the crowding-out mechanism[2]. In the following sections, we empirically test this theoretical prediction using our survey results.

3 Survey and Data

We designed two online surveys to examine the model's implications in the context of DLC participation: a discrete choice experiment in the first survey and a randomized incentive experiment in the second. Both surveys follow a similar three-module structure. The first part introduces the DLC program and presents a common background narrative intended to make the social visibility of participation salient. The second part contains the main experimental module: a discrete choice experiment (DCE) in the first survey or a randomized compensation experiment in the second survey. The final part collects respondents' demographic characteristics.

[2]See Figure 2 of Bénabou and Tirole (2006) and our Figure 6 for a visual illustration of the U-shaped relationship.

3.1 Survey Introduction

The theoretical model highlights how households' participation decisions may be shaped by their perceived social image within the community. To provide respondents with a concrete decision context, both surveys begin with the following common background narrative and a set of intrinsic motivation questions before the DCE or randomized incentive module.

At the beginning of the survey, respondents read the following introduction:

Background 1 *Suppose that your residential community introduces a peak-demand response program. During periods of high electricity demand, participating households may be asked to increase the temperature setting of their air conditioners by 1–2 degrees for a short period. This adjustment helps reduce pressure on the local power grid and lowers the risk of power outages.*

All participating households in the community are offered the same compensation schedule. This schedule is publicly known and applies equally to every participating household.

To provide residents with information about community participation in the program, the community periodically posts a list of participating households on a residential bulletin board located in a common area. The list is visible to other residents and indicates only whether each household participates in the program.

Please answer the following questions based on how you would evaluate and respond to this program.

This introduction places the participation decision in a concrete community setting. The common compensation schedule ensures that all households within the same residential community face an identical monetary incentive. Since the schedule is publicly known, other residents can infer the compensation associated with participation when they observe a household listed on the residential bulletin board. The publicly visible participation list and the common compensation schedule therefore jointly generate the signal (x, m) observed by neighbors in the model.

We additionally elicit three related dimensions of intrinsic motivation through the following questions. These questions are intended to make respondents' prosocial considerations more salient and to measure heterogeneity in public spirit, perceived social norms, and reputational concern:

Question 1 *(Public spirit)*

During periods of peak electricity demand, the local power grid faces substantial pressure. Increasing the temperature setting of your home air conditioner by 1–2 degrees for a short period can help reduce peak demand, ease pressure on the grid, and lower the risk of unexpected power outages. How likely would you be to make this adjustment in order to contribute to local grid reliability?

a. Very likely   b. Likely   c. Not likely   d. Not at all likely

Question 2 *(Social norm)*

Based on your own experience, please indicate how much you agree or disagree with the following statement:

Most people in my community are willing to make small personal sacrifices, such as limiting their air conditioner use during peak-demand periods, for collective benefits such as improving local grid reliability or protecting the environment.

a. Totally agree   b. Agree   c. Disagree   d. Strongly disagree

Question 3 *(Reputational concern)*

Suppose that a household in your residential community is listed as a participant in the peak-demand response program on the residential bulletin board. Under the program, all participating households receive the same monetary compensation for each peak-demand event.

How likely do you think it is that this household participates mainly because they want to contribute to local grid reliability instead of money?

a. Very likely   b. Likely   c. Not likely   d. Not at all likely

The first two questions capture public spirit and perceived social norms, two important non-monetary determinants of participation. Public spirit reflects an individual's active concern for collective welfare. Perceived social norms capture the extent to which individuals believe that prosocial behavior is common or socially expected in their community, particularly in relation to the behavior of their neighbors (Cialdini et al., 1990)[3].

The third question captures respondents' perceptions of the reputational implications of receiving monetary compensation. It asks whether respondents believe that other residents would still view a participating household as motivated primarily by a desire to support local grid reliability

[3] In our sample, the correlation between responses to the two questions is 0.282, suggesting that they capture related but distinct dimensions of prosocial motivation.

when compensation is provided. A lower response (less likely) indicates a stronger belief that monetary compensation obscures the household's prosocial motive and makes its participation appear more self-interested. We interpret lower responses as indicating a greater potential reputational cost associated with monetary compensation.

3.2 The Discrete Choice Experiment Design

After the common introduction and intrinsic-motivation questions, 775 respondents in the first survey group complete the DCE module. The DCE includes four attributes, as detailed in Table 1. The first is the minimum indoor AC temperature setpoint, with three levels: 27°C, 28°C, and 29°C. This range allows meaningful peak-load reduction without imposing unrealistically restrictive controls. Since 26°C is widely promoted in China as the minimum energy-saving standard, 27°C provides a practical baseline, while higher setpoints capture increasing comfort and decision-making costs (Heyes and Saberian, 2019; **?**).

The second attribute is the duration of each AC temperature restriction episode, with levels of 30, 120, and 240 minutes. The third attribute is the data-sharing policy, which is important to DLC design because appliance-level control requires household usage information and may raise privacy concerns (Frederiks et al., 2015; Stenner et al., 2017). We consider three levels for this attribute: (i) appliance usage data is used solely for control purposes and is not stored, (ii) data is stored by the electricity suppliers, and (iii) stored by the government. The government storage option is intended to reflect a scenario in which data use by electricity providers is constrained by regulatory oversight, potentially changing the privacy concerns (Kernstock et al., 2025). The final attribute is compensation, set at 50, 100, 200, and 400 CNY based on Zhao et al. (2022). Respondents are explicitly informed that compensation is a one-time payment for five load-control episodes during one summer.

Based on these attributes and levels, we use an orthogonal experimental design to generate a representative set of DLC alternatives (**??**)[4]. Although efficient designs are increasingly common in discrete choice experiments, they require prior parameter assumptions (Mariel et al., 2021). In this sense, the orthogonal design improves the transparency in our setting by showing that the

[4]We assess the orthogonality of the design ex post by computing the correlation matrix for the four attributes in Appendix Table A3. The highest correlation is 0.1, between compensation and temperature, while all other correlations are below 0.1 and close to zero.

estimated non-monotonic relationship is not shaped by prior parameter assumptions embedded in the experimental design.

Next, we pair every two generated alternatives into one choice card and add the status quo as the third alternative. We then drop cards with implausible combinations[5] and randomly select 16 cards from the remaining set to ensure plausible trade-offs (van den Broek-Altenburg and Atherly, 2020). An example card is shown in Table 2, and the full set is shown in Appendix Table A1.

An important feature of our DCE is its ranking format. As shown in Table 2, respondents rank all three alternatives instead of choosing only their most preferred option. This allows us to compare *each* DLC alternative directly with the status quo and construct indicators of DLC acceptance conditional on each DLC alternative's attributes[6]. This feature helps identify the U-shaped relationship between DLC acceptance and monetary incentives since each respondent evaluates multiple DLC alternatives with different attribute combinations, reducing the concern that the estimated relationship is driven by one particular combination[7].

Although the ranking format helps reduce bias from preferences for particular attribute combinations, an important concern remains. In principle, including a zero-compensation level in the DCE would allow us to measure the baseline no-compensation acceptance rate in the absence of monetary incentives, which should be higher if small monetary incentives crowd out intrinsic motivation. However, we do not include this zero-compensation level since the appearance of compensation as an attribute may already alter respondents' decision frame. Once compensation is presented as an attribute in the choice tasks, respondents may evaluate every alternative through a monetary frame, including an alternative with zero compensation. They may interpret zero compensation as an insufficiently low payment instead of as a purely voluntary opportunity to contribute to local grid reliability. As a result, a zero-compensation level embedded in the DCE may underestimate true acceptance in the absence of monetary incentives and distort the elicited preferences. Prior studies document related forms of within-card and across-card contamination in stated choice experiments, such as zero-price effects (Shampanier et al., 2007; Adamowicz et al.,

[5] We assume that higher temperature setpoints, longer episode durations, and data storage by any entity reduce utility. Therefore, comparisons such as 29°C for 240 minutes with 50 CNY versus 27°C for 30 minutes with 400 CNY are implausible and are dropped.

[6] By comparison, a traditional DCE that records only the most preferred alternative does not always reveal whether a respondent prefers the second-ranked DLC alternative to the status quo. See Section 4 for details.

[7] Compared with the randomized experiment discussed later, where each respondent evaluates only one representative DLC alternative, the ranking-based DCE covers a wider range of DLC combinations within the same survey.

2011; Tjiong et al., 2026) and ordering effects (Boxebeld, 2024).

To address this concern and identify the benchmark acceptance rate in the absence of monetary incentives, we rely on a second randomized experiment. Respondents are randomly assigned to different incentive frames. In the voluntary no-compensation group, they are asked whether they would participate in the DLC program without any compensation, and no monetary information is shown on the interface. This design provides an alternative benchmark for assessing whether low compensation reduces participation relative to voluntary participation without compensation.

3.3 The Randomized Experiment Design

In the second survey group, we sampled 647 new respondents to provide randomized incentives. Each respondent is shown one representative DLC alternative: the household minimum AC temperature setpoint is 28°C for 60 minutes per episode, up to five times during the summer. Respondents are then randomly assigned to one of five incentive groups: voluntary and no compensation, 50 CNY, 100 CNY, 200 CNY, or 400 CNY compensation, which follow the attribute levels in the DCE to supplement the no-compensation participation.

The survey question is shown below. Respondents report their willingness to participate on a four-point scale:

Question 4 *During peak-demand periods between 18:30 and 23:00 in July and August, the electricity supplier will directly control the AC at your home if necessary. Your AC temperature setpoint cannot be set below 28°C. This restriction will last for 60 minutes for each control. The electricity suppliers will notify you at 30 minutes before the control. The program will be activated five times over the summer. The one-time compensation is X. How likely would you be to participate in this AC-control program?*

a. Very likely b. Likely c. Not likely d. Not at all likely

where X corresponds to no compensation, 50 CNY, 100 CNY, 200 CNY, or 400 CNY compensation, depending on the randomly assigned group.

The randomized experiment also allows us to revisit the concern that making compensation salient in the DCE may itself affect respondents' stated preferences. After completing the DCE or randomized-experiment module, respondents answer a follow-up question asking whether insufficient compensation is a main barrier to accepting DLC. Appendix Figure E1 shows that respondents in the DCE sample are significantly more likely than those in the voluntary no-compensation group to identify low compensation as a barrier to DLC participation. This finding suggests that the DCE framework increases the salience of monetary incentives, supporting our concern that a zero-compensation alternative embedded in the DCE would not provide a clear benchmark for purely voluntary participation.

3.4 Data Collection and Descriptive Statistics

We conducted the survey online in two major Chinese cities, Beijing and Shanghai, in early 2021. Beijing and Shanghai are among the most economically developed cities in China and have relatively high shares of well-educated residents. These residents are more likely to understand and engage with innovative mechanisms in the electricity market (Balta-Ozkan et al., 2013). At the same time, Beijing is about 5°C cooler than Shanghai on average in July and August. This variation allows us to examine how climatic conditions shape preferences for DLC attributes with implications for policy design.

The survey was administered through WenJuanXing, a leading Chinese online survey platform that provides professional sampling services. Participants were drawn from the platform's nationwide pool of verified users. Respondents participated voluntarily, completed the questionnaire online, and received a small fixed payment from the platform after being informed of the study's academic purpose and their right to withdraw. After validation, the final DCE sample includes 370 respondents from Beijing and 405 from Shanghai.

We report basic demographic characteristics for the full sample in Table 3 and provide subgroup statistics in Appendix B. First, the average respondent in our sample was about 34 years old in 2021, compared with a median age of 42 in the 2020 city census. Second, women account for about 63 percent of our sample, compared with roughly 49 percent in the census. Third, more than 96 percent of respondents hold a bachelor's degree or higher, far above the citywide average of about 40 percent. Finally, respondents have lived in their respective cities for about 20 years on average, suggesting familiarity with local norms, climate, and community dynamics.

This younger and better-educated sample is relevant for studying DLC. DLC programs in China are still at a pilot stage, and most residents are unfamiliar with their technical and institutional features. Sampling relatively young and well-educated respondents helps ensure that participants understand the mechanisms and implications of DLC and can answer the choice questions reliably (Balta-Ozkan et al., 2013; Nilsson and Bartusch, 2024). Moreover, existing AC DLC pilots, such as the CoolNYC program in the US, generally require participants to use smartphone-based interfaces. Younger and more technologically fluent individuals are therefore a relevant group of likely early adopters (Henriksen et al., 2025).

The third column of Table 3 reports t-statistics from balance tests comparing the DCE sample with the pooled randomized-experiment sample. Most observed characteristics are balanced across the two samples. The only difference statistically significant at the 10 percent level is the share of respondents living in homes built between 2000 and 2010. The final column reports F-statistics from ANOVA balance tests across the five randomized incentive groups. Only the indicator for double-layer windows differs significantly across groups at the 5 percent level, indicating that the random assignment achieved broad balance.

Table 4 reports descriptive statistics for the intrinsic value measures from Questions 1 to 3. Most respondents report relatively high levels of agreement with the values. Therefore, in the empirical analysis, we convert the prosocial value measures into binary indicators to avoid small sample sizes for less frequently chosen response levels.

4 Empirical Strategy

In this section, we describe our empirical strategy. We first present the logit models used to analyze the discrete choice experiment results. We then introduce the regression models for the randomized experiment. Finally, we describe the estimation strategy for the theoretical model parameters.

4.1 Random-Parameter Logit and Willingness to Accept

We use the RPL model as the main specification to estimate preferences and WTA in the DCE. This estimation will contribute to the DLC pricing in the Chinese context. Each choice card is indexed by j, and each respondent i faces three alternatives on each card, indexed by k. The respondent is assumed to choose the alternative that yields the highest utility[8]. We define a binary variable y_{ijk} to indicate the observed choice outcome:

$$y_{ijk} = 1\{\text{alternative } k \text{ is top ranked for respondent } i \text{ on choice card } j\}$$

To model how y_{ijk} depends on the attributes of each alternative, we define the following specification for latent utility u_{ijk}:

$$u_{ijk} = x'_{ijk}\beta + \varepsilon_{ijk}$$

where u_{ijk} represents the utility that respondent i obtains from alternative k on choice card j. β is the coefficient vector, which is interpreted as the marginal change of indirect utility. ε_{ijk} is the error term. x_{ijk} is a vector of binary indicators for alternative-specific attributes that includes:

$$x_{ijk} = (temp_{28}, temp_{29}, dur_{120}, dur_{240}, company, gov, compen, ASC)'_{ijk}$$

where *temp* denotes the temperature setpoint, and *dur* denotes the duration indicators. The subscripts indicate the corresponding attribute levels. *company* and *gov* indicate whether appliance-use information is stored by the electricity supplier or by the government, respectively. *compen* is the compensation amount, which is treated as a continuous variable for the WTA estimation[9]. The omitted levels serve as the base categories. *ASC* is an indicator for the status quo alternative and captures the utility of remaining outside the DLC program. We calculate the WTA for each attribute as follows:

$$-\frac{\beta_{attribute}}{\beta_{compen}} \tag{5}$$

where $\beta_{attribute}$ is the coefficient on one of the non-monetary attribute indicators.

For each choice card j, the respondent determines their choice of alternatives that can realize

[8] Although respondents rank all three alternatives in the survey, the RPL model uses only the top-ranked alternative because its purpose is to recover preference parameters from direct comparisons among all alternatives.

[9] The RPL model relies on a continuous compensation specification for WTA estimation, as WTA is recovered from the marginal rate of substitution between non-monetary attributes and monetary compensation (Fox et al., 2012). This WTA specification should be distinguished from the test for the U-shaped relationship below, where compensation enters through categorical indicators to examine whether DLC acceptance responds non-monotonically to monetary incentives.

the highest latent utility level:

$$y_{ijk} = 1\{u_{ijk} \geq u_{ijk'} \text{ for all } k' \neq k\}$$

We assume that ε_{ijk} follows the type-I extreme value distribution and allow β to vary across respondents according to the distribution $f(\beta|\omega)$. The log-likelihood function can then be written as:

$$l^{\mathrm{RPL}}(\omega) = \sum_{i,j,k} 1\{y_{ijk} = 1\} \ln \int_{\beta} \frac{\exp(x'_{ijk}\beta)}{\sum_m \exp(x'_{ijm}\beta)} f(\beta|\omega) d\beta \tag{6}$$

where ω denotes the parameters of the distribution f. We assume that f is a normal distribution, with $\omega = (\mu, \sigma^2)$ representing the mean and variance of β. A statistically significant estimate of σ^2 indicates heterogeneity in respondents' preferences over the DLC attributes.

We further estimate an unrestricted RPL model that allows the coefficients to vary by city to examine between-city heterogeneity:

$$u_{ijk} = (x_{ijk} \cdot 1\{city_i = bj\})'\beta_{bj} + (x_{ijk} \cdot 1\{city_i = sh\})'\beta_{sh} + \varepsilon_{ijk} \tag{7}$$

where $city_i$ denotes the survey city of respondent i, with bj indicating Beijing and sh indicating Shanghai.

4.2 Crowding-Out Effects in the Discrete Choice Experiment

We exploit the DCE ranking format to test for the U-shaped relationship between DLC acceptance and monetary compensation. Since our interest is whether respondents are willing to accept DLC, we define the dependent variable as an indicator for whether each DLC alternative is ranked above the status quo:

$$y_{ijk} = 1\{\text{alternative } k \text{ is ranked above the status quo by respondent } i \text{ on choice card } j\} \tag{8}$$

where $k = 3$ (the status quo alternative) is excluded since it cannot be compared with itself. Appendix Table A2 reports the distribution of this dependent variable across choice cards, showing that acceptance rates exceed 50 percent for most of them.

To examine the crowding-out effects, we replace the continuous compensation measure with categorical compensation indicators and estimate the following regression:

$$y_{ijk} = \alpha_{100} \cdot compen_{ijk}^{100} + \alpha_{200} \cdot compen_{ijk}^{200} + \alpha_{400} \cdot compen_{ijk}^{400} + attribute'_{ijk}\beta + x'_i\gamma + \eta_{ijk} \quad (9)$$

where *compen* denotes binary indicators for the respective compensation levels, with 50 CNY omitted as the base category. α_j is interpreted as the difference in DLC acceptance between alternatives offering j CNY and those offering 50 CNY in compensation. $attribute_{ijk}$ is a vector of the remaining DCE attributes, excluding *compen* and ASC. x_i is a vector of control variables including respondent characteristics. If the U-shaped relationship holds, we would expect $\alpha_j < 0$ for small j's[10].

4.3 Crowding-Out Effects in the Randomized Incentives

Using the randomized experiment sample, we estimate the following regression to test for crowding-out effects:

$$y_i = \beta_{50}b_i^{50} + \beta_{100}b_i^{100} + \beta_{200}b_i^{200} + \beta_{400}b_i^{400} + x'_i\gamma + \xi_i \quad (10)$$

where y_i is a binary variable indicating whether respondent i is willing to participate in the provided DLC program[11]. Binary variables $b_i^j = 1$ if respondent i is offered j CNY. The voluntary no-compensation group is omitted as the base group. β_j measures the difference in DLC participation between respondents offered j CNY in compensation and those offered no compensation. x_i is a vector of control variables that includes characteristics with statistically significant differences across groups at the 10 percent level or below, and ξ_i is the error term. If the U-shaped relationship holds, we would expect $\beta_j < 0$ for small j's.

4.4 Estimation of Model Parameters

The randomized experiment design allows us to construct a maximum likelihood estimator for the parameters of the theoretical model and to simulate the implied U-shaped relationship. This

[10] Since 50 CNY instead of 0 CNY is the base category, α_j measures the difference in DLC acceptance between alternatives offering j CNY and those offering 50 CNY. Therefore, a negative α_j requires the crowding-out persists beyond 50 CNY. We argue that this condition holds as we empirically find a negative α_{100} in Section 5.

[11] $y_i = 1$ if respondent i chooses *Very likely* or *Likely* to Question 4.

estimation provides supplementary evidence on the crowding-out mechanism, since Proposition 1 shows that crowding-out arises only for certain parameter values.

We use the analytical solution of contribution x in equation (3) to construct the likelihood function. The main estimation challenge is that the continuous contribution x in the theoretical model is observed as a binary participation decision in our survey. To address this issue, we draw on the latent-variable framework. Specifically, the observed binary choice x_i for respondent i is defined as follows:

$$x_i = 1\{x_i^* \geq 0\}$$

where x_i is the observed binary choice in the survey, and latent x_i^* is the analytical solution from the theoretical model in equation (3).

Substituting the right-hand side of equation (3) for x_i^*, the log-likelihood function can be written as:

$$l(\bar{v}, \Omega, \mu) = \sum_i \{x_i \ln \mathbb{P}(x_i = 1|m_i) + (1 - x_i) \ln[1 - \mathbb{P}(x_i = 1|m_i)]\} \tag{11}$$

where

$$\mathbb{P}(x_i = 1|m_i) = \Phi\left(\frac{(\bar{v}_s + \bar{v}_m m_i)/k + \mu_s \rho(m_i) - \mu_m \chi(m_i)}{\sqrt{(\sigma_s^2 + m_i^2 \sigma_m^2 + 2m_i \sigma_{sm})/k^2}}\right)$$

where Φ denotes the CDF of the standard normal distribution. We provide the detailed derivation in Appendix D. We estimate $\bar{v}$, Ω, and μ by maximizing the log-likelihood function in equation (11) with the observed m_i and x_i.

5 Empirical Results

5.1 Attribute Preferences and Willingness to Accept

We begin with the baseline RPL model using the full sample and the specification (6) with continuous compensation to estimate DLC preferences and WTA. Table 5 presents the estimation results. Columns (1) and (2) report the estimated means and standard deviations of the random coefficients, respectively.

In Column (1), higher temperature setpoints significantly reduce the indirect utility, with coefficients below -1.0, which is consistent with the increased comfort cost of stricter temperature restrictions (Heyes and Saberian, 2019). Longer control durations lower indirect utility, and the coefficients for 120 and 240 minutes are similar (both coefficients between -0.7 and -0.6). This suggests that the marginal disutility of episode duration may decline as duration increases. Yilmaz et al. (2022) similarly document limited sensitivity to longer control durations among some groups for heat pumps and electric vehicles. Appliance-use data storage also reduces the indirect utility relative to no data storage. The estimates are similar whether the data are stored by the electricity supplier or by the government (both coefficients around -0.5). This suggests that respondents are concerned about appliance-use data being stored regardless of the storage entity, possibly reflecting broader concerns about data privacy or the enforcement of data protection regulations in DLC (Stenner et al., 2017; Familia and Horne, 2022; Marikyan et al., 2024). The positive coefficient on monetary compensation indicates that higher payments increase indirect utility under the specification with a continuous compensation term. Finally, the ASC coefficient is below -2.4 and statistically significant at the 1 percent level, indicating that in general the status quo yields much lower utility than the DLC alternatives[12].

Column (2) reports the estimated standard deviations of the random coefficients. The significant estimates for most attributes indicate substantial heterogeneity in respondent preferences, except for the 28°C setpoint and government data storage. Column (3) reports the WTA estimates for each attribute level based on specification (5). Respondents require substantially higher compensation for temperature restrictions than for other attributes, with WTA estimates about two to four times larger. By contrast, the WTA estimates for control duration are much lower. These results are consistent with the interpretation above that respondents may be less sensitive to additional episode duration once episodes become sufficiently long.

We now turn to the unrestricted model in specification (7), which allows the coefficients to vary by city. Table 6 presents the results. Columns (1) and (4) report the mean estimates for Beijing and Shanghai, respectively. Columns (3) and (6) report the WTA for the two cities. Respondents in Shanghai have larger negative coefficients on control duration, which may reflect the city's warmer summer climate. Respondents in Beijing have a larger negative coefficient on data storage, potentially reflecting differences in institutional trust or privacy concerns between the two cities

[12] This negative estimate is consistent with the high DLC acceptance rate in the randomized experiment. See Section 5.4 and Figure 3.

(Good et al., 2017).

In Columns (2) and (5), fewer attributes have statistically significant standard deviation estimates. This suggests that part of the preference heterogeneity in Table 5 is explained by between-city differences. This finding highlights the importance of accounting for local conditions in DLC program design.

5.2 Factors Contributing to DLC Acceptance and the Role of Prosocial Values

Before examining crowding-out effects in the randomized experiment, we first analyze how DLC acceptance is associated with respondent characteristics.

We pool the DCE sample from both cities and regress the DLC acceptance indicator in the DCE defined by specification (8) on respondents' personal, household, and housing characteristics, controlling for DLC attributes. The results are shown in Figure 1. Among household characteristics, higher household education is positively associated with DLC acceptance. This may reflect that respondents from better-educated households have a clearer understanding of the benefits of DLC for local power grids, consistent with prior evidence that limited knowledge can act as a barrier to DSM participation (Nilsson and Bartusch, 2024). The share of vulnerable household members (under age 6 or over age 65) is negatively associated with DLC acceptance, likely reflecting care-giving concerns in the family, although the estimate is not statistically significant at the 5 percent level. By contrast, longer residence in the survey city is positively associated with DLC acceptance, which may reflect a stronger sense of community attachment and greater willingness to contribute to local public goods (Kowalska-Pyzalska, 2018). Among housing characteristics, better insulation quality and newer buildings are both positively associated with DLC acceptance. This is consistent with the argument that better thermal performance of the dwelling reduces the comfort cost of DLC (Sridhar et al., 2024).

More importantly, after controlling for monetary compensation in the DLC alternatives, we examine the role of intrinsic motivation and find that stronger intrinsic motivation is positively associated with DLC acceptance in Figure 1. This result is consistent with the prior studies emphasizing prosocial drivers of DSM participation (Gołębiowska et al., 2020; Hackbarth and Löbbe, 2020). Appendix Table B5 provides further evidence by extending the RPL model with interaction terms between the ASC and the intrinsic value measures. Since the ASC captures the utility of the status quo relative to DLC alternatives, the negative interaction estimates (from -2.9 to -2.2) indicate that intrinsic motivation increases the utility of taking DLC.

It is important to note that the results in Figure 1 and Appendix Table B5 do not contradict the crowding-out effects implied by the theoretical model. The crowding-out mechanism concerns how *changes* in extrinsic incentives alter the role of prosocial motivation in decision-making. By contrast, the analyses above and the related literature show that, holding extrinsic incentives fixed, respondents with stronger prosocial motivation are still more willing to accept DLC. This is an important distinction emphasized by the literature (Bénabou and Tirole, 2006; Mellström and Johannesson, 2008). Therefore, policymakers should not conclude that prosocial motivation is irrelevant simply because monetary incentives may crowd it out. Instead, our results suggest that promoting prosocial values and strengthening the broader prosocial environment remain important tools for encouraging DLC and DSM participation, as discussed in the prior literature (Hackbarth and Löbbe, 2020; Gołębiowska et al., 2020; Williams et al., 2023).

5.3 Crowding-Out Effects in the Discrete Choice Experiment

In this section, we examine crowding-out effects in DLC acceptance using the pooled DCE sample from Beijing and Shanghai. We test for a U-shaped relationship between monetary incentives and DLC acceptance by estimating specification (9), which relates acceptance to categorical compensation levels. The estimates are presented in the first panel of Figure 2[13]. Relative to the 50 CNY compensation level, 100 CNY significantly reduces the likelihood of DLC acceptance, with an estimated coefficient of approximately -0.2. The coefficient becomes positive at 200 CNY, at about 0.1, and increases further to approximately 0.4 at 400 CNY. These estimates are consistent with a U-shaped relationship between DLC acceptance and monetary compensation in the DCE.

To assess whether the U-shaped relationship reflects the crowding-out of intrinsic motivation, we split respondents by their intrinsic motivation measures and re-estimate specification (9). The remaining panels of Figure 2 report the estimates separately by public spirit, perceived social norms, and reputational concern.

We find that the negative coefficient on the 100 CNY compensation level is concentrated among

[13]The figure reports estimates from regressions that control for respondent characteristics. Full estimates are reported in Appendix Table C1; estimates without controls are reported in Appendix Table C2.

respondents with stronger prosocial values or greater reputational concern. By contrast, the negative effect at 100 CNY largely disappears among respondents with weaker prosocial values or lower reputational concern[14]. This result is consistent with crowding-out as the initial decline in acceptance is concentrated among respondents whose participation is more likely to be driven by intrinsic motivation.

5.4 Crowding-Out Effects in the Randomized Incentives and Heterogeneity

In this section, we use the randomized experiment sample from both Beijing and Shanghai to provide additional evidence on the crowding-out effect. Figure 3 shows responses to Question 4 across the five randomized groups. Overall participation rates are high and exceed 40 percent in all groups, consistent with the negative estimated ASC coefficient in the RPL models, which suggests that respondents are generally more likely to choose a DLC alternative against the status quo. Across groups, participation rates are lower in the 50 CNY to 200 CNY groups than in the voluntary no-compensation group, while the rate is higher in the 400 CNY group, suggesting the presence of crowding-out effects.

We estimate the regression specification (10). The control variables include baseline characteristics that differ across the randomized groups at the 10 percent significance level or below in the ANOVA tests reported in the final column of Table 3. The first panel of Figure 4 presents the estimation results[15].

The estimates support the presence of crowding-out from extrinsic incentives. Relative to voluntary participation without compensation, offering 50 CNY or 100 CNY reduces DLC participation, with estimated coefficients of approximately -0.5 and -0.9, respectively. When compensation increases to 200 CNY, the estimate rises relative to the 100 CNY level and returns to a magnitude close to that at 50 CNY. At 400 CNY, the estimated effect becomes positive, suggesting that monetary incentives become sufficiently large to outweigh the crowding-out effect in the participation decision.

Next, as in the DCE analysis, we examine whether the U-shaped relationship in the randomized experiment is driven by the crowding-out of prosocial motivation. The remaining panels of Figure

[14] Appendix Table C4 reports Wald tests of between-group differences. The estimates differ significantly across groups at the 5 percent level for the 100 CNY compensation level.

[15] The full estimates are reported in Appendix Table C3.

4 report estimates separately by respondents' attitudes toward each intrinsic motivation measure. The negative effects of low compensation are again concentrated among respondents with stronger prosocial values or greater reputational concern. Among respondents with positive prosocial attitudes or higher reputational concern, the coefficients on the 50 to 200 CNY compensation levels are all negative. By contrast, among respondents with weaker prosocial values or lower reputational concern, these coefficients are substantially smaller in magnitude and close to zero[16]. This finding suggests that crowding-out is weaker among respondents with limited intrinsic motivation, for whom there is less prosocial motivation to be crowded out.

Given the baseline results of crowding-out effects, we examine heterogeneity by respondent characteristics in Figure 5, which reports the estimated effects of the 50 to 200 CNY compensation treatments relative to the voluntary no-compensation group. The largest heterogeneity appears across housing characteristics. Respondents living in homes without insulation layers or in buildings constructed before 2000 exhibit more negative responses to the 50 to 200 CNY compensation levels. These households may face greater comfort costs from AC control because of poorer thermal performance. Monetary compensation may therefore make the private burden of participation more salient, leading to larger declines in DLC participation among households with less energy-efficient housing.

At the same time, older respondents and long-term local residents exhibit weaker crowding-out effects, possibly because they are more embedded in the local community (Perlaviciute and Steg, 2014). Through repeated interactions, their prosocial orientation may already be partly known to other residents. As a result, accepting a small payment may be less likely to generate reputational concerns.

The estimated crowding-out effects vary little by gender or household education. This finding differs from some prior evidence. For example, Mellström and Johannesson (2008) find that crowding-out in blood donation is stronger among women. The absence of gender differences in our setting may reflect the context-specific nature of crowding-out in DLC participation.

As an additional robustness check, Appendix Figure E2 examines heterogeneity by household income. This analysis provides a suggestive test of the insult-effect interpretation, under which a small payment may reduce participation because respondents perceive it as an offensively low val-

[16] Wald tests are reported in Appendix Table C5.

uation of their contribution, instead of crowding out intrinsic motivation (Gneezy and Rustichini, 2000). If this interpretation were the primary mechanism, the negative response to low compensation should be substantially stronger among higher-income households. However, the estimated effects of the 50 to 200 CNY compensation levels are not more negative for the high-income group. This finding suggests that the negative response to low compensation is not limited to higher-income respondents and is unlikely to be fully explained by an insult effect.

5.5 Structural Estimation of the Theoretical Model

Finally, we estimate the structural parameters of the theoretical model. Using MLE proposed in Section 4.4, the parameter estimates are reported in Appendix Table D1. Based on these estimates, we simulate the average DLC participation rates and present them in Figure 6. The curve labeled baseline illustrates a U-shaped relationship between monetary incentives and DLC participation.

To examine how intrinsic and extrinsic motivation jointly shape aggregate DLC participation, we conduct a series of counterfactual simulations that vary the variance of individuals' preferences for monetary gains σ_m^2. This parameter captures heterogeneity, or noise, in monetary motivation across households. When σ_m^2 is larger, declining a small payment becomes a more informative signal that an individual is not primarily motivated by monetary gain. This strengthens the reputational cost associated with accepting low compensation and, in turn, intensifies crowding-out effects (Bénabou and Tirole, 2006). The dashed curves in Figure 6 present these counterfactual results. Consistent with this intuition, increasing σ_m^2 strengthens crowding-out, as shown by the red and green curves. Conversely, reducing σ_m^2, as illustrated by the orange curve, weakens crowding-out and raises overall DLC participation.

5.6 Comparative Discussion and Cost-Benefit Analysis

To conclude the empirical analysis, we compare our DCE-based WTA estimates with existing evidence on DSM programs and conduct a cost–benefit analysis to assess whether the DCE-implied compensation requirements are feasible for electricity suppliers.

First, we summarize several DLC valuation studies from the United States, Switzerland, and Australia in Table 7. To improve comparability across distinct climatic and economic contexts, we focus on appliance-level load control in the residential sector. Overall, DLC participation rates are generally high (above 50 percent) for programs involving AC, heat pumps, or electric vehicle charging. The compensation required for representative DLC attributes typically ranges from 10 to 60 percent of the average monthly household electricity bill, indicating substantial cross-study variation. In our RPL estimation, the implied WTA for different temperature or duration levels represents approximately 19.1 to 60.6 percent of a typical Chinese household's monthly summer electricity bill (assuming 350 CNY per month). This range is broadly consistent with the international evidence and findings from existing DLC trials.

Second, we conduct a rough cost–benefit analysis from the electricity supplier's perspective to assess the feasibility of our DLC design. We assume that the supplier's costs under the DLC program are the compensation paid to participating households, while the benefits come from avoided investment in electricity storage capacity needed to cover summer peak–off-peak gaps[17]. For illustration, we consider an extreme scenario that requires the highest compensation with a 29°C setpoint and a four-hour control window. Engineering rules of thumb suggest that raising a typical household AC's setpoint from 26°C to 29°C reduces cooling electricity use by approximately 0.36–0.60 kWh per episode. If such control is activated five times over the summer, each participating household would save roughly 1.8–3.0 kWh. Based on benchmark investment costs for grid-forming battery storage systems in China, approximately 370–500 CNY per kWh,[18] this demand reduction corresponds to an avoided storage investment of about 660–1,500 CNY per household per summer. This amount substantially exceeds the estimated WTA in Table 5.

6 Conclusions and Policy Implications

This study investigates the role of extrinsic and intrinsic incentives in shaping households' decisions to participate in a DLC program in AC control, based on the survey conducted in Beijing and Shanghai. In the first survey group, we uncover respondents' preferences over the DLC attributes and estimate respondents' WTA compensation using a discrete choice experiment and the RPL model. We reveal a U-shaped relationship between DLC acceptance and monetary incentives, highlighting the potential crowding-out effects in DLC programs.

[17]This is a lower bound of the benefits. If the economic losses from power outages are taken into account, the potential benefits would be much larger.

[18]We use quoted prices from a bid for energy storage capacity by the Power Construction Corporation of China on December 4, 2024.

In the second survey group, we employ a randomized experiment that varies the levels of extrinsic compensation to test for crowding-out effects. Our findings show that small extrinsic incentives can diminish intrinsic motivation, resulting in lower participation rates compared to voluntary participation without compensation. Conversely, sufficiently large monetary incentives can reverse this effect, as extrinsic rewards begin to outweigh reputational concerns and dominate the decision-making.

These findings offer implications for incentive design in energy policy. First, policymakers should leverage behavioral data to segment households based on their intrinsic motivation levels (Gołębiowska et al., 2020; Williams et al., 2023). Tailored messaging and differentiated incentives, for example, emphasizing community impact for prosocial households and offering competitive compensation for more economically driven ones, can enhance program effectiveness and cost-efficiency.

Second, uniform compensation may not be optimal. Small incentives risk crowding-out prosocial participation without significantly boosting enrollment from financially motivated households. Policymakers should consider tiered or conditional compensation schemes, where rewards increase based on participation frequency, community enrollment thresholds, or performance-based criteria. This suggestion is consistent with simulation-based findings in Sridhar et al. (2024) and Golmaryami et al. (2024).

Third, the presence of crowding-out effects highlights the importance of aligning incentives with prevailing social norms. Monetary incentives may be less effective, or even counterproductive, when they make participation appear transactional and impose reputational costs on participants. Policymakers should therefore assess how different rewards are perceived and prioritize incentive-compatible designs that are also norm-compatible, such as bill credits, public recognition, or rewards framed as compensation for contributing to grid reliability. This implication is also consistent with our heterogeneity analysis and with Graf et al. (2023) who show that incentives are more effective when they are perceived as normatively appropriate.

References

Adamowicz, W., Dupont, D., Krupnick, A., and Zhang, J. (2011). Valuation of cancer and microbial disease risk reductions in municipal drinking water: An analysis of risk context using multiple valuation methods. *Journal of Environmental Economics and Management*, 61(2):213–226.

Ali, S. N. and Bénabou, R. (2020). Image versus Information: Changing Societal Norms and Optimal Privacy. *American Economic Journal: Microeconomics*, 12(3):116–164.

Anwar, M. B. and O'Malley, M. (2021). Strategic Participation of Residential Thermal Demand Response in Energy and Capacity Markets. *IEEE Transactions on Smart Grid*, 12(4):3070–3085.

Bakare, M. S., Abdulkarim, A., Zeeshan, M., and Shuaibu, A. N. (2023). A comprehensive overview on demand side energy management towards smart grids: challenges, solutions, and future direction. *Energy Informatics*, 6(1):4.

Balta-Ozkan, N., Davidson, R., Bicket, M., and Whitmarsh, L. (2013). Social barriers to the adoption of smart homes. *Energy Policy*, 63:363–374.

Bartling, B., Weber, R. A., and Yao, L. (2015). Do Markets Erode Social Responsibility? *. *The Quarterly Journal of Economics*, 130(1):219–266.

Battigalli, P. and Dufwenberg, M. (2022). Belief-Dependent Motivations and Psychological Game Theory. *Journal of Economic Literature*, 60(3):833–882.

Bender, J., Fait, L., and Wetzel, H. (2024). Acceptance of demand-side flexibility in the residential heating sector — Evidence from a stated choice experiment in Germany. *Energy Policy*, 191:114145.

Bernheim, B. D. (1994). A Theory of Conformity. *Journal of Political Economy*, 102(5):841–877.

Blanke, J., Beder, C., and Klepal, M. (2017). An Integrated Behavioural Model towards Evaluating and Influencing Energy Behaviour—The Role of Motivation in Behaviour Demand Response. *Buildings*, 7(4):119.

Boxebeld, S. (2024). Ordering effects in discrete choice experiments: A systematic literature review across domains. *Journal of Choice Modelling*, 51:100489.

Bradley, P., Coke, A., and Leach, M. (2016). Financial incentive approaches for reducing peak electricity demand, experience from pilot trials with a UK energy provider. *Energy Policy*, 98:108–120.

Brennan, N. and van Rensburg, T. M. (2023). Does intermittency management improve public acceptance of wind energy? A discrete choice experiment in Ireland. *Energy Research & Social Science*, 95:102917.

Bénabou, R. and Tirole, J. (2006). Incentives and Prosocial Behavior. *American Economic Review*, 96(5):1652–1678.

Cialdini, R. B., Reno, R. R., and Kallgren, C. A. (1990). A focus theory of normative conduct: Recycling the concept of norms to reduce littering in public places. *Journal of Personality and Social Psychology*, 58(6):1015–1026.

d'Adda, G. (2011). Motivation crowding in environmental protection: Evidence from an artefactual field experiment. *Ecological Economics*, 70(11):2083–2097.

d'Adda, G., Dufwenberg, M., Passarelli, F., and Tabellini, G. (2020). Social norms with private values: Theory and experiments. *Games and Economic Behavior*, 124:288–304.

Daniel, A. M., Persson, L., and Sandorf, E. D. (2018). Accounting for elimination-by-aspects strategies and demand management in electricity contract choice. *Energy Economics*, 73:80–90.

Dugstad, A., Grimsrud, K. M., Kipperberg, G., Lindhjem, H., and Navrud, S. (2021). Scope Elasticity of Willingness to pay in Discrete Choice Experiments. *Environmental and Resource Economics*, 80(1):21–57.

D'Ettorre, F., Banaei, M., Ebrahimy, R., Pourmousavi, S. A., Blomgren, E. M. V., Kowalski, J., Bohdanowicz, Z., Łopaciuk Gonczaryk, B., Biele, C., and Madsen, H. (2022). Exploiting demand-side flexibility: State-of-the-art, open issues and social perspective. *Renewable and Sustainable Energy Reviews*, 165:112605.

Fabianek, P., Liepold, C., and Madlener, R. (2025). A multi-criteria assessment framework for direct load control in residential buildings from an occupants' perspective. *Energy*, 321:135456.

Familia, T. and Horne, C. (2022). Customer trust in their utility company and interest in household-level battery storage. *Applied Energy*, 324:119772.

Faruqui, A. and Sergici, S. (2010). Household response to dynamic pricing of electricity: a survey of 15 experiments. *Journal of Regulatory Economics*, 38(2):193–225.

Fischer, P. and Huddart, S. (2008). Optimal Contracting with Endogenous Social Norms. *American Economic Review*, 98(4):1459–1475.

Fox, J. T., Kim, K. I., Ryan, S. P., and Bajari, P. (2012). The random coefficients logit model is identified. *Journal of Econometrics*, 166(2):204–212.

Frederiks, E. R., Stenner, K., and Hobman, E. V. (2015). Household energy use: Applying behavioural economics to understand consumer decision-making and behaviour. *Renewable and Sustainable Energy Reviews*, 41:1385–1394.

Frey, B. S., Oberholzer-Gee, F., and Eichenberger, R. (1996). The Old Lady Visits Your Backyard: A Tale of Morals and Markets. *Journal of Political Economy*, 104(6):1297–1313.

Gamma, K., Mai, R., Cometta, C., and Loock, M. (2021). Engaging customers in demand response programs: The role of reward and punishment in customer adoption in Switzerland. *Energy Research & Social Science*, 74:101927.

Georgarakis, E., Bauwens, T., Pronk, A.-M., and AlSkaif, T. (2021). Keep it green, simple and socially fair: A choice experiment on prosumers' preferences for peer-to-peer electricity trading in the Netherlands. *Energy Policy*, 159:112615.

Gleue, M., Unterberg, J., Löschel, A., and Grünewald, P. (2021). Does demand-side flexibility reduce emissions? Exploring the social acceptability of demand management in Germany and Great Britain. *Energy Research & Social Science*, 82:102290.

Gneezy, U. and Rustichini, A. (2000). Pay Enough or Don't Pay at All*. *The Quarterly Journal of Economics*, 115(3):791–810.

Goff, S. H., Waring, T. M., and Noblet, C. L. (2017). Does Pricing Nature Reduce Monetary Support for Conservation?: Evidence From Donation Behavior in an Online Experiment. *Ecological Economics*, 141:119–126.

Golmaryami, S., Nunes, M. L., and Ferreira, P. (2024). The role of social learning on consumers' willingness to engage in demand-side management: An agent-based modelling approach. *Smart*

Energy, 14:100138.

Good, N., Ellis, K. A., and Mancarella, P. (2017). Review and classification of barriers and enablers of demand response in the smart grid. *Renewable and Sustainable Energy Reviews*, 72:57–72.

Gołębiowska, B., Bartczak, A., and Czajkowski, M. (2020). Energy Demand Management and Social Norms. *Energies*, 13(15):3779.

Graf, C., Suanet, B., Wiepking, P., and Merz, E.-M. (2023). Social norms offer explanation for inconsistent effects of incentives on prosocial behavior. *Journal of Economic Behavior & Organization*, 211:429–441.

Gyamfi, S. and Krumdieck, S. (2011). Price, environment and security: Exploring multi-modal motivation in voluntary residential peak demand response. *Energy Policy*, 39(5):2993–3004.

Hackbarth, A. and Löbbe, S. (2020). Attitudes, preferences, and intentions of German households concerning participation in peer-to-peer electricity trading. *Energy Policy*, 138:111238.

Hall, N. L., Jeanneret, T. D., and Rai, A. (2016). Cost-reflective electricity pricing: Consumer preferences and perceptions. *Energy Policy*, 95:62–72.

Hanley, N., Mourato, S., and Wright, R. E. (2001). Choice Modelling Approaches: A Superior Alternative for Environmental Valuatioin? *Journal of Economic Surveys*, 15(3):435–462.

Henriksen, I. M., Strömberg, H., Branlat, J., Diamond, L., Garzon, G., Kuch, D., Yilmaz, S., and Motnikar, L. (2025). The role of gender, age, and income in demand-side management acceptance: A literature review. *Energy Efficiency*, 18(3):17.

Heyes, A. and Saberian, S. (2019). Temperature and Decisions: Evidence from 207,000 Court Cases. *American Economic Journal: Applied Economics*, 11(2):238–265.

Kaczmarski, J., Jones, B., and Chermak, J. (2022). Determinants of Demand Response Program Participation: Contingent Valuation Evidence from a Smart Thermostat Program. *Energies*, 15(2):590.

Kernstock, P., Harms, C., Hein, A., and Krcmar, H. (2025). Establishing and governing data ecosystems at the crossroads of centralization and decentralization. *Electronic Markets*, 35(1):71.

Khanna, N. Z., Guo, J., and Zheng, X. (2016). Effects of demand side management on Chinese household electricity consumption: Empirical findings from Chinese household survey. *Energy Policy*, 95:113–125.

Kim, K., Choi, J., Lee, J., Lee, J., and Kim, J. (2023). Public preferences and increasing acceptance of time-varying electricity pricing for demand side management in South Korea. *Energy Economics*, 119:106558.

Kowalska-Pyzalska, A. (2018). What makes consumers adopt to innovative energy services in the energy market? A review of incentives and barriers. *Renewable and Sustainable Energy Reviews*, 82:3570–3581.

Ladenburg, J., Jensen, K. L., Lodahl, C., and Keles, D. (2022). Testing for non-linear willingness to accept compensation for controlled electricity switch-offs using choice experiments. *Energy*, 238:121749.

Lashmar, N., Wade, B., Molyneaux, L., and Ashworth, P. (2022). Motivations, barriers, and enablers for demand response programs: A commercial and industrial consumer perspective. *Energy Research & Social Science*, 90:102667.

Lynch, M. A., Nolan, S., Devine, M. T., and O'Malley, M. (2019). The impacts of demand response participation in capacity markets. *Applied Energy*, 250:444–451.

Mariel, P., Hoyos, D., Meyerhoff, J., Czajkowski, M., Dekker, T., Glenk, K., Jacobsen, J. B., Liebe, U., Olsen, S. B., Sagebiel, J., and Thiene, M. (2021). *Environmental Valuation with Discrete Choice Experiments: Guidance on Design, Implementation and Data Analysis*. SpringerBriefs in Economics. Springer International Publishing, Cham.

Marikyan, D., Papagiannidis, S., Rana, O. F., and Ranjan, R. (2024). General data protection regulation: a study on attitude and emotional empowerment. *Behaviour & Information Technology*, 43(14):3561–3577.

Mellström, C. and Johannesson, M. (2008). Crowding Out in Blood Donation: Was Titmuss Right? *Journal of the European Economic Association*, 6(4):845–863.

Meyer, E. L., Overen, O. K., Obileke, K., Botha, J. J., Anderson, J. J., Koatla, T. A. B., Thubela, T., Khamkham, T. I., and Ngqeleni, V. D. (2021). Financial and economic feasibility of bio-digesters for rural residential demand-side management and sustainable development. *Energy Reports*, 7:1728–1741.

Ming, Z., Song, X., Mingjuan, M., Lingyun, L., Min, C., and Yuejin, W. (2013). Historical review of demand side management in China: Management content, operation mode, results assessment and relative incentives. *Renewable and Sustainable Energy Reviews*, 25:470–482.

Mohanty, S., Panda, S., Parida, S. M., Rout, P. K., Sahu, B. K., Bajaj, M., Zawbaa, H. M., Kumar, N. M., and Kamel, S. (2022). Demand side management of electric vehicles in smart grids: A survey on strategies, challenges, modeling, and optimization. *Energy Reports*, 8:12466–12490.

Morrissey, K., Plater, A., and Dean, M. (2018). The cost of electric power outages in the residential sector: A willingness to pay approach. *Applied Energy*, 212:141–150.

Newsham, G. R. and Bowker, B. G. (2010). The effect of utility time-varying pricing and load control strategies on residential summer peak electricity use: A review. *Energy Policy*, 38(7):3289–3296.

Nilsson, A. and Bartusch, C. (2024). Empowered or enchained? Exploring consumer perspectives on Direct Load Control. *Energy Policy*, 192:114248.

Oerlemans, L. A. G., Chan, K.-Y., and Volschenk, J. (2016). Willingness to pay for green electricity: A review of the contingent valuation literature and its sources of error. *Renewable and Sustainable Energy Reviews*, 66:875–885.

Panda, S., Mohanty, S., Rout, P. K., Sahu, B. K., Bajaj, M., Zawbaa, H. M., and Kamel, S. (2022). Residential Demand Side Management model, optimization and future perspective: A review. *Energy Reports*, 8:3727–3766.

Perlaviciute, G. and Steg, L. (2014). Contextual and psychological factors shaping evaluations and acceptability of energy alternatives: Integrated review and research agenda. *Renewable and Sustainable Energy Reviews*, 35:361–381.

Roldán-Blay, C., Escrivá-Escrivá, G., and Roldán-Porta, C. (2019). Improving the benefits of demand response participation in facilities with distributed energy resources. *Energy*, 169:710–718.

Scharnhorst, L., Sloot, D., Lehmann, N., Ardone, A., and Fichtner, W. (2024). Barriers to demand response in the commercial and industrial sectors – An empirical investigation. *Renewable and Sustainable Energy Reviews*, 190:114067.

Shampanier, K., Mazar, N., and Ariely, D. (2007). Zero as a Special Price: The True Value of Free Products. *Marketing Science*, 26(6):742–757.

Shipman, R., Gillott, M., and Naghiyev, E. (2013). SWITCH: Case Studies in the Demand Side Management of Washing Appliances. *Energy Procedia*, 42:153–162.

Skoczkowski, T., Bielecki, S., Wołowicz, M., Sobczak, L., Węglarz, A., and Gilewski, P. (2024). Participation in demand side response. Are individual energy users interested in this? *Renewable Energy*, 232:121104.

Sloot, D., Lehmann, N., and Ardone, A. (2022). Explaining and promoting participation in demand response programs: The role of rational and moral motivations among German energy consumers. *Energy Research & Social Science*, 84:102431.

Sridhar, A., Honkapuro, S., Ruiz, F., Stoklasa, J., Annala, S., and Wolff, A. (2024). Residential consumer enrollment in demand response: An agent based approach. *Applied Energy*, 374:123988.

Sridhar, A., Honkapuro, S., Ruiz, F., Stoklasa, J., Annala, S., Wolff, A., and Rautiainen, A. (2023). Toward residential flexibility—Consumer willingness to enroll household loads in demand response. *Applied Energy*, 342:121204.

Steinhorst, J., Klöckner, C. A., and Matthies, E. (2015). Saving electricity – For the money or the environment? Risks of limiting pro-environmental spillover when using monetary framing. *Journal of Environmental Psychology*, 43:125–135.

Stenner, K., Frederiks, E. R., Hobman, E. V., and Cook, S. (2017). Willingness to participate in direct load control: The role of consumer distrust. *Applied Energy*, 189:76–88.

Tjiong, J., Dekker, T., Hess, S., Giergiczny, M., Ojeda-Cabral, M., and Czajkowski, M. (2026). Capturing zero-price effects in stated choice surveys: implications for willingness-to-pay and welfare. *Journal of Choice Modelling*, 58:100582.

Uddin, M., Romlie, M. F., Abdullah, M. F., Abd Halim, S., Abu Bakar, A. H., and Chia Kwang, T. (2018). A review on peak load shaving strategies. *Renewable and Sustainable Energy Reviews*, 82:3323–3332.

van den Broek-Altenburg, E. and Atherly, A. (2020). Using discrete choice experiments to measure preferences for hard to observe choice attributes to inform health policy decisions. *Health Economics Review*, 10(1):18.

Vohs, K. D., Mead, N. L., and Goode, M. R. (2006). The Psychological Consequences of Money. *Science*, 314(5802):1154–1156.

Wen, C., Steadman, S., Rafaq, M. S., Vatougiou, P., and Deakin, M. (2025). Can reduction of local carbon emissions motivate participation in demand-side flexibility programs? Evidence from the

United Kingdom. *Applied Energy*, 388:125610.

Williams, B., Bishop, D., Gallardo, P., and Chase, J. G. (2023). Demand Side Management in Industrial, Commercial, and Residential Sectors: A Review of Constraints and Considerations. *Energies*, 16(13):5155.

Xu, X., Chen, C.-f., Zhu, X., and Hu, Q. (2018). Promoting acceptance of direct load control programs in the United States: Financial incentive versus control option. *Energy*, 147:1278–1287.

Yilmaz, S., Chanez, C., Cuony, P., and Patel, M. K. (2022). Analysing utility-based direct load control programmes for heat pumps and electric vehicles considering customer segmentation. *Energy Policy*, 164:112900.

Yilmaz, S., Cuony, P., and Chanez, C. (2021). Prioritize your heat pump or electric vehicle? Analysing design preferences for Direct Load Control programmes in Swiss households. *Energy Research & Social Science*, 82:102319.

Zha, D., Yang, G., Wang, W., Wang, Q., and Zhou, D. (2020). Appliance energy labels and consumer heterogeneity: A latent class approach based on a discrete choice experiment in China. *Energy Economics*, 90:104839.

Zhang, S., Jiao, Y., and Chen, W. (2017). Demand-side management (DSM) in the context of China's on-going power sector reform. *Energy Policy*, 100:1–8.

Zhao, Y., Yu, W., Guo, D., and He, X. (2022). Household Customers' Cost of Interruption: Survey Study on Summer Electricity Peak of Xi'an City. *Frontiers in Energy Research*, 9:802018.

Zhao, Z., Yu, C., Yew, M., and Liu, M. (2015). Demand side management: A green way to power Beijing. *Journal of Renewable and Sustainable Energy*, 7(4):041505.

Zhou, K. and Yang, S. (2015). Demand side management in China: The context of China's power industry reform. *Renewable and Sustainable Energy Reviews*, 47:954–965.

Tables

Table 1: Attributes and Levels in the Discrete Choice Experiment

Attributes	Illustration	Levels	Variable label
AC temperature restriction	The minimum temperature (Celsius) that can be set by the household	Not lower than 27 degrees Not lower than 28 degrees Not lower than 29 degrees	(base) $temp_{28}$ $temp_{29}$
AC control duration	How long the AC temperature restriction is effective	Last for 30 minutes Last for 120 minutes Last for 240 minutes	(base) dur_{120} dur_{240}
Information storage	Where the home appliance use data is stored	Data used for control only and do not store By electricity suppliers By government	(base) *company* *gov*
Monetary compensation	How much money is offered to household if they accept the alternative including controls for five times	50 CNY 100 CNY 200 CNY 400 CNY	*compen*

Note: The last column reports the variable abbreviations used in the regression tables.

Table 2: Example of a Choice Card

Please **rank** the following three alternatives (A, B, and C) from most preferred to least preferred. The compensation below is a one-time payment for the total of **five** control events in July and August from 18:30 to 23:00.			
	Alternative A	**Alternative B**	**Alternative C**
You can set your home AC temperature **not lower than**	28 degrees	27 degrees	Status quo
Duration of above temperature restriction	30 minutes	4 hours	
Who will store your home appliance use data	Electricity suppliers	Government	
Compensation you will be paid	50 CNY	50 CNY	
Rank the three alternatives	( )	( )	( )

Table 3: Descriptive Statistics of Demographic Characteristics

Variables	Full sample mean	Full sample std. err.	DCE versus RCT t-statistics	RCT group balance F-statistics
Respondent's characteristics				
Age in 2021	34.89	13.23	-1.58	0.32
0-24 years old	0.19	0.39	1.01	0.72
25-39 years old	0.57	0.50	-1.53	0.46
40-65 years old	0.15	0.36	0.54	0.04
Above 65 years old	0.08	0.28	-0.79	0.30
Female	0.63	0.48	1.48	0.37
Education level				
College graduate	0.96	0.19	0.87	0.15
Years that live in the survey city	20.07	14.50	-0.46	2.48*
0-5 years	0.24	0.43	0.48	1.28
6-15 years	0.22	0.41	0.74	2.02
16-30 years	0.25	0.43	-0.84	2.22*
Above 30 years	0.29	0.45	-0.97	1.63
Household characteristics				
Number of college graduates	1.64	1.45	-0.71	0.21
Household member age				
Number of members below 6	0.34	0.52	0.40	0.62
Number of members in 6-18	0.36	0.60	-0.32	1.31
Number of members in 18-65	2.46	1.04	0.65	0.60
Number of members >65	0.35	0.70	-0.34	0.31
Household monthly total income				
0-4,000 CNY	0.03	0.17	1.26	0.04
4,000-7,500 CNY	0.06	0.24	1.07	0.52
7,500-10,000 CNY	0.31	0.46	0.14	1.03
10,000-15,000 CNY	0.29	0.46	-0.53	0.56
15,000-30,000 CNY	0.22	0.41	0.14	0.19
Above 30,000 CNY	0.09	0.29	-0.37	0.39
Housing characteristics				
House area (m^2)	98.45	33.03	-0.53	0.01
Equipped with double-layer windows	0.48	0.50	-0.39	3.03**
Equipped with insulation layers	0.54	0.50	0.71	0.51
House built year				
House built before 2000	0.34	0.47	-1.00	2.39*
House built in 2000-2010	0.43	0.50	1.86*	2.19*
House built after 2010	0.23	0.42	-1.17	0.71
N	1,422			

Note: The full sample pools respondents from the DCE and randomized experiment (RCT) samples in both cities. The t-statistics comparing the DCE and randomized-experiment samples are based on tests of mean differences between the DCE sample and the pooled randomized-experiment sample. The F-statistics comparing randomized-experiment groups are based on ANOVA tests across the five incentive groups. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Descriptive statistics by specific groups and city are reported in Appendix Tables B1 to B3.

Table 4: Descriptive Statistics of Prosocial Values

Variables	Full sample mean	Full sample std. err.	Between DCE and RCT t-statistics	Between RCT groups F-statistics
Public spirit (Question 1): more prosocial if more likely				
Very likely	0.28	0.45	1.02	0.39
Likely	0.63	0.48	-1.48	0.57
Not likely	0.08	0.28	0.38	0.34
Not at all likely	0.01	0.09	1.75*	1.78
Social norm (Question 2): stronger perceived social norm if greater agreement				
Totally agree	0.14	0.34	-0.96	0.64
Agree	0.58	0.49	0.08	0.76
Disagree	0.26	0.44	-1.22	0.75
Strongly disagree	0.02	0.14	-1.13	1.80
Reputational concern (Question 3): higher concern if less likely				
Not at all likely	0.28	0.45	1.23	1.27
Not likely	0.57	0.50	-1.31	0.74
Likely	0.13	0.34	-0.88	1.29
Very likely	0.02	0.14	-0.32	0.18
N	1,422			

Note: The full sample pools respondents from the DCE and randomized experiment (RCT) samples in both cities. The t-statistics comparing the DCE and randomized-experiment samples are based on tests of mean differences between the DCE sample and the pooled randomized-experiment sample. The F-statistics comparing randomized-experiment groups are based on ANOVA tests across the five incentive groups. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Descriptive statistics by specific groups and city are reported in Appendix Table B4.

Table 5: Estimates of DLC Attribute Preferences: Full Sample

	RPL		Pricing (WTA)
	Mean	Std. dev.	
	(1)	(2)	(3)
Temperature restriction (base: 27 degrees)			
$temp = 28$	-1.165***	0.190	166.43
	(0.107)	(0.371)	
$temp = 29$	-1.961***	1.206***	280.14
	(0.154)	(0.193)	
Control duration (base: 30 mins)			
$dur = 120$	-0.601***	0.563***	85.86
	(0.096)	(0.200)	
$dur = 240$	-0.663***	0.634***	94.71
	(0.114)	(0.216)	
Information storage (base: no storage)			
company	-0.497***	0.621***	71.00
	(0.100)	(0.187)	
gov	-0.554***	0.279	79.14
	(0.116)	(0.333)	
Monetary compensation			
compen	0.007***	0.008***	
	(0.001)	(0.001)	
ASC			
ASC	-2.458***	3.531***	
	(0.217)	(0.254)	
N	37,200		
Log likelihood	-2683		

Note: This table reports RPL estimates from specification (6) using the full sample. Column (3) reports the implied pricing schedule (WTA) for each attribute level, calculated based on specification (5). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

Table 6: Estimates of DLC Attribute Preferences: by City

Sample:	Beijing			Shanghai		
	Mean	Std. dev.	Pricing (WTA)	Mean	Std. dev.	Pricing (WTA)
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature restriction (base: temperature at 27 degrees)						
$temp = 28$	-1.115***	0.172	159.29	-1.190***	0.543*	170.00
	(0.141)	(0.530)		(0.132)	(0.285)	
$temp = 29$	-2.021***	1.123***	288.71	-1.864***	1.145***	266.29
	(0.214)	(0.301)		(0.174)	(0.232)	
Control duration (base: control for 30 mins)						
$dur = 120$	-0.270**	0.642**	38.57	-0.862***	0.333	123.14
	(0.130)	(0.303)		(0.128)	(0.280)	
$dur = 240$	-0.491***	0.454	70.14	-0.873***	0.229	124.71
	(0.154)	(0.318)		(0.137)	(0.218)	
Information storage (base: no information share)						
company	-0.615***	0.585*	87.86	-0.257**	0.047	36.71
	(0.150)	(0.322)		(0.113)	(0.250)	
gov	-0.659***	0.441	94.14	-0.228*	0.010	32.57
	(0.177)	(0.407)		(0.131)	(0.240)	
Monetary compensation						
compen	0.007***	0.005***		0.007***	0.013***	
	(0.001)	(0.001)		(0.001)	(0.001)	
ASC						
ASC	-3.380***	4.648***		-1.416***	0.513***	
	(0.450)	(0.523)		(0.144)	(0.131)	
N			37,200			
Log likelihood			-2221			

Note: This table reports RPL estimates from specification (7) by allowing the coefficients to vary by city. Columns (3) and (6) report the implied pricing schedule (WTA) for each attribute level based on specification (5). $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

Table 7: Comparison with the Existing Literature

	Contents of DLC	Estimated pricing	Share on monthly elec. bill
Xu et al. (2018)	Adjust AC within 3°F from 2 to 5 pm in summer	30 USD leads to 66% participation rate	30/178 = 16.9%
Yilmaz et al. (2021)	Restrict heat pump use for 1.5 hours in winter	71.4 CHF from once per week to 3 times per week	71.4/326 = 21.9%
Yilmaz et al. (2021)	Restrict electric vehicle charge for 1.5 hours in peak months	27.9 CHF from once per week to 3 times per week	27.9/326 = 8.6%
Ausgrid CoolSaver (See Ausgrid website)	Restrict AC for 1-2 hours and 4-6 times per summer	100 AUD leads to 87% participation	100/165 = 60.3%

Note: The last column is calculated using average monthly household electricity bills in each country. Bills are based on data from the US EIA, Swissgrid, and the Australian Energy Regulator.

Figures

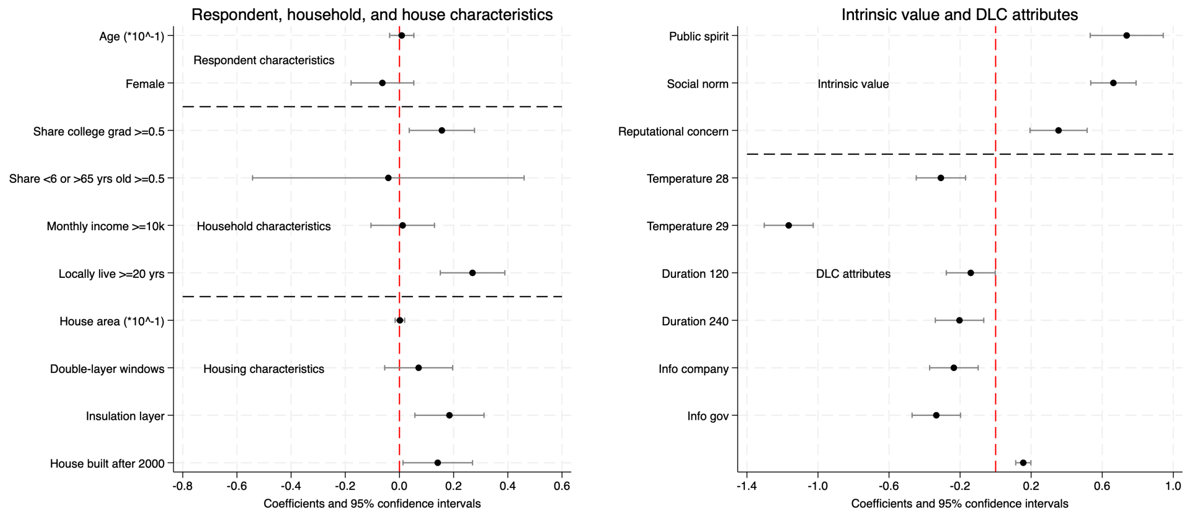

Figure 1: Logit Estimates of DLC Acceptance on Characteristics

Note: This figure reports logit estimates and 95% confidence intervals of DLC acceptance defined by (8) on respondents' personal, household, and housing characteristics by using the DCE sample and controlling for DLC attributes. Some variables labeled in the figure are divided by 10 to improve the readability.

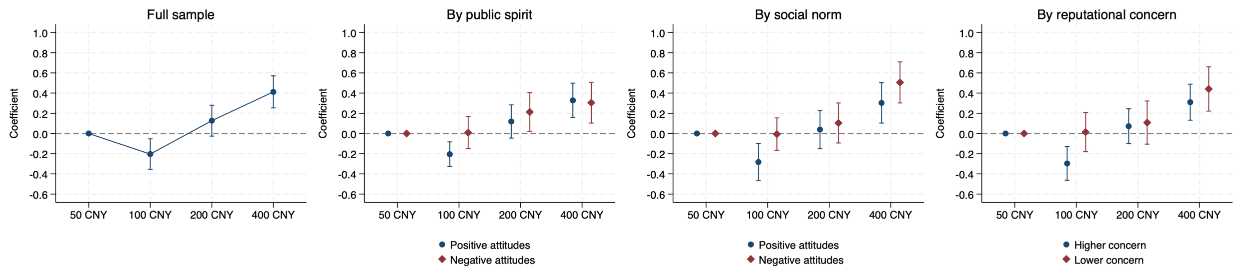

Figure 2: Crowding-Out in the DCE

Note: This figure reports the estimated coefficients and 95% confidence intervals from the DCE regression with categorical compensation in specification (9) by controlling for demographic characteristics. The 50 CNY level is the base category. The first plot uses the full sample; the remaining plots split respondents by intrinsic-value attitudes at the midpoint of the four responses. Full estimates are reported in Appendix Table C1. Wald tests are reported in Appendix Table C4. Estimates without control variables are reported in Appendix Table C2.

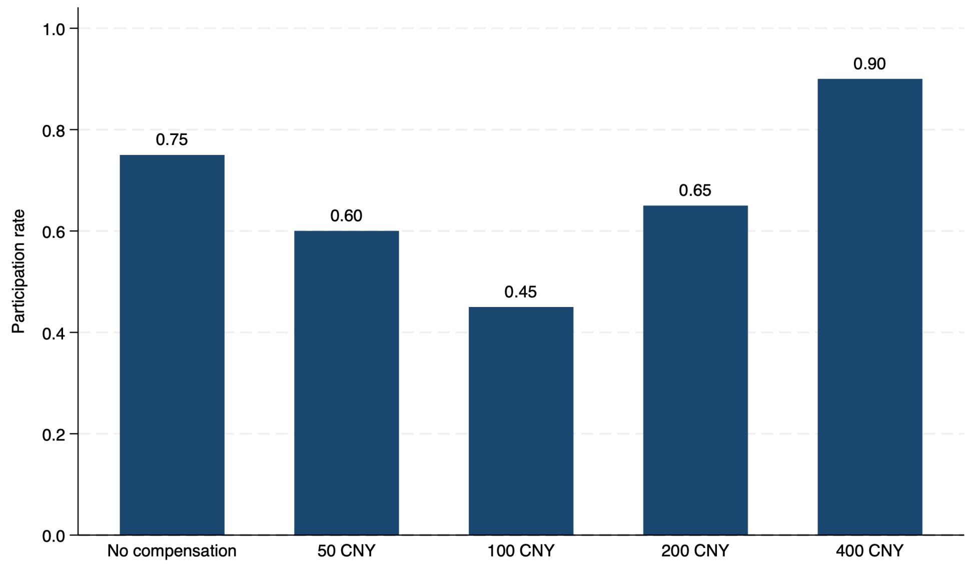

Figure 3: Participation Rates by Incentive Condition in the Randomized Experiment

Note: This figure shows participation rates by randomized incentive condition. Participation equals one if the respondent chooses Very likely or Likely in Question 4. No compensation refers to the voluntary and no-compensation group.

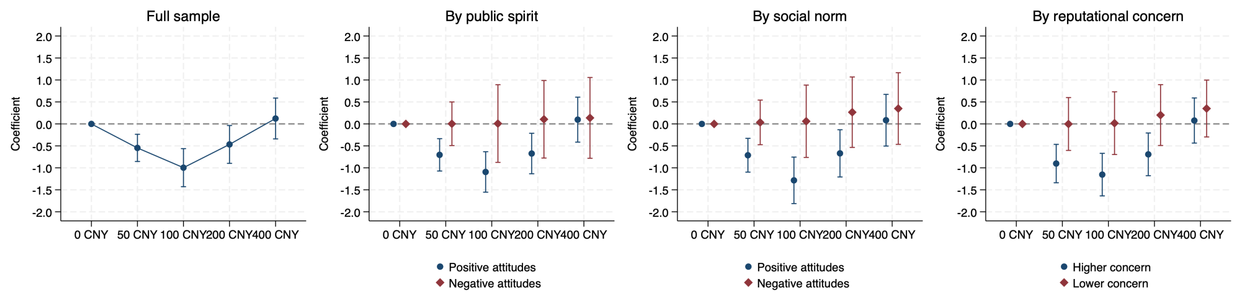

Figure 4: Crowding-Out in the Randomized Experiment

Note: This figure reports the estimated coefficients and 95% confidence intervals using the randomized experiment in specification (10) by controlling for variables significantly different across groups at or below the 10 percent level. The no-compensation group (0 CNY) is the base category. The first plot uses the full sample; the remaining plots split respondents by intrinsic-value attitudes at the midpoint of the four responses. Full estimates are reported in Appendix Table C3. Wald tests are reported in Appendix Table C5.

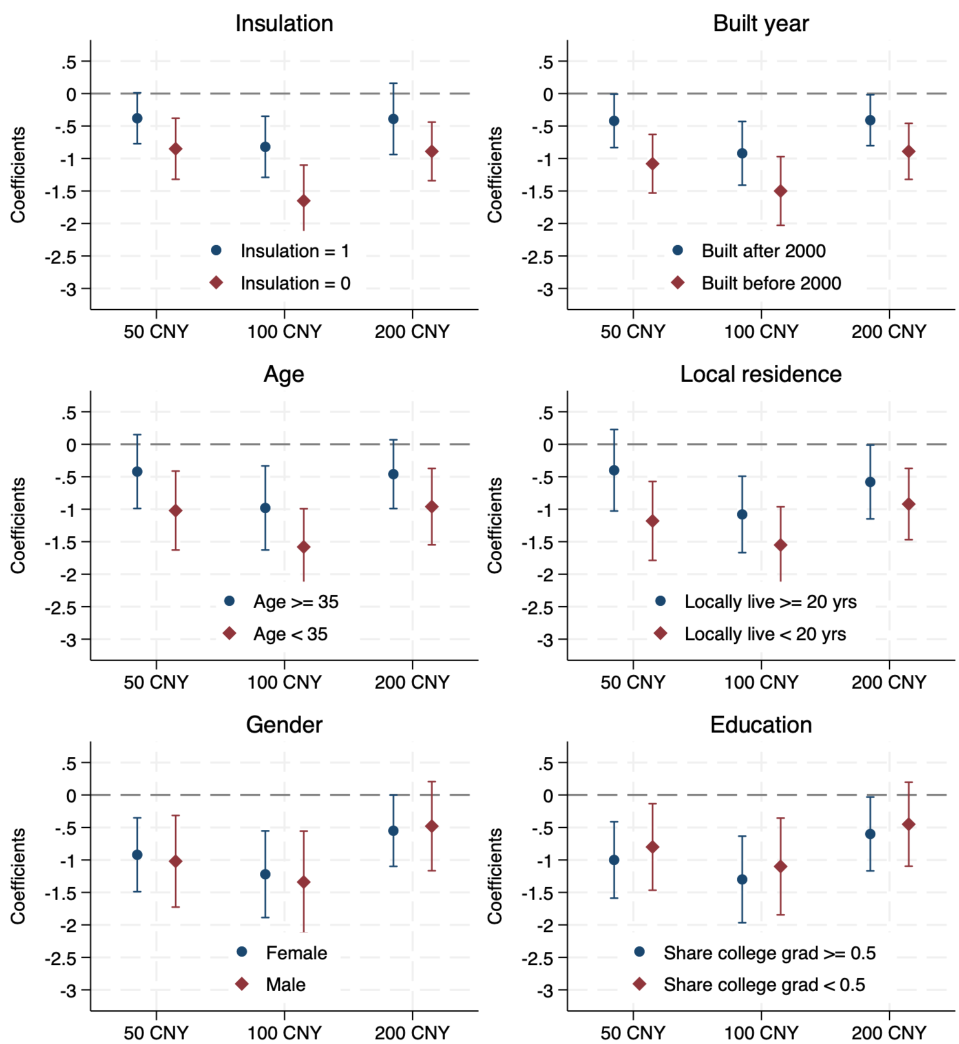

Figure 5: Heterogeneity Analysis of Crowding-Out in the Randomized Experiment

Note: This figure repeats the full sample regression in Figure 4 by further splitting the sample according to respondent characteristics at the median for continuous variables. We report only the coefficients and their 95% confidence intervals on 50 to 200 CNY to highlight the crowding-out effects. The base is the voluntary no-compensation group. The full estimates are reported in Appendix Table C6 to C8.

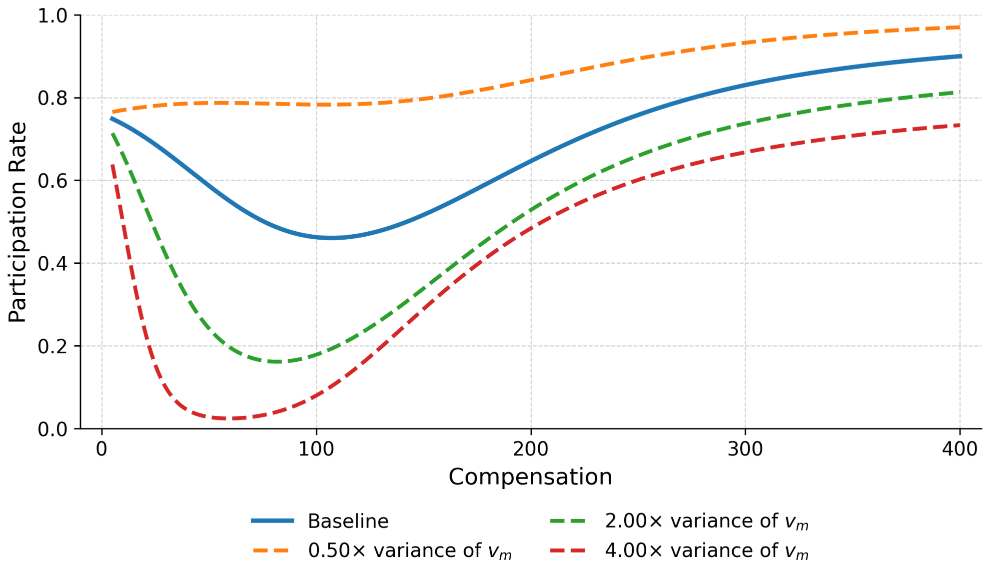

Figure 6: Simulated and Counterfactual Average DLC Participation Rates with Monetary Incentives

Note: This figure plots the simulated relationship between the acceptance rate and monetary compensation, along with three counterfactual experiments. The solid curve is based on the MLE estimates reported in Appendix Table D1 from the log-likelihood function in specification (11). The dashed curves show counterfactual simulations that vary σ_m^2.

A Choice Card List

Table A1 provides the full list of choice cards used in our survey.

Table A1: List of Choice Cards

Card number	Alternative A				Alternative B			
1	27 degrees	30 mins	no storage	50 CNY	29 degrees	30 mins	gov.	100 CNY
2	27	30	no	100	27	120	company	200
3	27	240	company	100	29	240	no	200
4	27	30	no	200	28	30	no	400
5	27	240	gov.	50	29	120	no	50
6	28	120	no	100	27	30	no	50
7	28	240	no	400	27	30	no	200
8	28	30	no	400	27	30	no	100
9	29	120	no	50	28	30	company	50
10	29	30	gov.	100	27	240	company	100
11	27	120	company	200	28	240	no	400
12	27	120	gov.	400	29	30	company	400
13	28	30	company	50	27	240	gov.	50
14	29	240	no	200	28	120	no	100
15	29	30	company	400	28	30	gov.	200
16	28	30	gov.	200	27	120	gov.	400

Note: This table lists all choice cards in the survey by group. For Alternative A and B, the first column reports the temperature setting, the second the control duration, the third the data-storage entity, and the fourth the compensation amount.

Table A2: DLC Acceptance Rate Relative to the Status Quo

Card number	Alternative A against status quo		Alternative B against status quo	
	Compensation	Participation rate	Compensation	Participation rate
1	50 CNY	0.76	100 CNY	0.57
2	100	0.73	200	0.63
3	100	0.63	200	0.47
4	200	0.81	400	0.72
5	50	0.70	50	0.47
6	100	0.68	50	0.76
7	400	0.71	200	0.73
8	400	0.82	100	0.76
9	50	0.48	50	0.56
10	100	0.51	100	0.65
11	200	0.69	400	0.64
12	400	0.72	400	0.61
13	50	0.67	50	0.69
14	200	0.55	100	0.68
15	400	0.66	200	0.72
16	200	0.71	400	0.77

Note: This table reports the rate at which each DLC alternative is preferred to the status quo in the DCE, corresponding to the dependent variable defined by specification (8).

Table A3: Correlation Matrix of DCE Attributes

	Duration	Information	Compensation
Temperature	-0.070	-0.070	0.104
Duration		0.000	0.000
Information			0.000

Note: This table reports the pairwise correlations among attribute levels in the alternatives generated by the orthogonal design.

B Additional Descriptive Statistics and Empirical Results

Table B1: Descriptive Statistics by Group and City: Respondent

Variables	DCE	0 CNY	50 CNY	100 CNY	200 CNY	400 CNY	Beijing	Shanghai
Age in 2021	34.65	34.16	34.76	35.31	35.29	35.10	34.10	35.61
	(13.10)	(12.55)	(13.63)	(14.00)	(12.81)	(13.20)	(13.14)	(13.29)
0–24 years old	0.20	0.19	0.22	0.21	0.16	0.18	0.22	0.17
	(0.40)	(0.39)	(0.41)	(0.41)	(0.37)	(0.38)	(0.42)	(0.37)
25–39 years old	0.57	0.59	0.55	0.54	0.59	0.58	0.54	0.60
	(0.50)	(0.49)	(0.50)	(0.50)	(0.49)	(0.49)	(0.50)	(0.49)
40–65 years old	0.15	0.16	0.15	0.15	0.16	0.15	0.16	0.15
	(0.36)	(0.37)	(0.36)	(0.36)	(0.37)	(0.36)	(0.37)	(0.36)
Above 65 years old	0.08	0.07	0.08	0.10	0.09	0.09	0.08	0.09
	(0.27)	(0.25)	(0.28)	(0.29)	(0.28)	(0.29)	(0.26)	(0.29)
Female	0.65	0.58	0.60	0.65	0.63	0.61	0.63	0.64
	(0.48)	(0.49)	(0.49)	(0.48)	(0.48)	(0.49)	(0.48)	(0.48)
Education level								
College graduate	0.96	0.97	0.96	0.96	0.96	0.96	0.97	0.96
	(0.20)	(0.18)	(0.20)	(0.19)	(0.20)	(0.19)	(0.18)	(0.21)
Years in the survey city	20.10	19.21	18.36	20.45	22.08	21.25	17.19	22.70
	(14.50)	(14.35)	(14.71)	(14.18)	(14.61)	(14.40)	(14.39)	(14.12)
0–5 years	0.24	0.22	0.29	0.23	0.22	0.23	0.28	0.20
	(0.43)	(0.42)	(0.46)	(0.42)	(0.41)	(0.42)	(0.45)	(0.40)
6–15 years	0.22	0.28	0.23	0.19	0.19	0.20	0.28	0.17
	(0.42)	(0.45)	(0.42)	(0.39)	(0.39)	(0.40)	(0.45)	(0.37)
16–30 years	0.27	0.24	0.21	0.32	0.25	0.28	0.22	0.28
	(0.45)	(0.43)	(0.41)	(0.47)	(0.43)	(0.45)	(0.42)	(0.45)
Above 30 years	0.27	0.26	0.28	0.27	0.35	0.29	0.22	0.35
	(0.45)	(0.44)	(0.45)	(0.45)	(0.48)	(0.45)	(0.42)	(0.48)
N	775	127	129	128	130	133	678	743

Note: This table reports descriptive statistics for respondent characteristics by randomized-experiment group and city. The DCE column pools all DCE respondents across both cities. The 0 to 400 CNY columns correspond to the five randomized-experiment groups and pool respondents across both cities. The Beijing and Shanghai columns pool the DCE and randomized-experiment samples within each city. Standard errors in parentheses.

Table B2: Descriptive Statistics by Group and City: Household

Variables	DCE	0 CNY	50 CNY	100 CNY	200 CNY	400 CNY	Beijing	Shanghai
Number of college graduates	1.61	1.57	1.68	1.66	1.64	1.65	1.60	1.67
	(1.38)	(1.29)	(1.43)	(1.64)	(1.43)	(1.46)	(1.47)	(1.43)
Household member age								
Number of members <6	0.33	0.31	0.32	0.33	0.38	0.34	0.34	0.33
	(0.52)	(0.50)	(0.52)	(0.53)	(0.54)	(0.53)	(0.52)	(0.52)
Number of members in 6–18	0.34	0.29	0.38	0.40	0.38	0.36	0.36	0.37
	(0.58)	(0.53)	(0.65)	(0.60)	(0.60)	(0.59)	(0.60)	(0.59)
Number of members in 18–65	2.46	2.52	2.46	2.47	2.38	2.44	2.48	2.43
	(1.04)	(1.00)	(1.09)	(1.03)	(1.04)	(1.05)	(1.06)	(1.02)
Number of members >65	0.35	0.33	0.39	0.33	0.37	0.36	0.35	0.36
	(0.71)	(0.68)	(0.77)	(0.64)	(0.70)	(0.71)	(0.70)	(0.70)
Household monthly total income								
0–4,000 CNY	0.04	0.03	0.03	0.03	0.03	0.03	0.05	0.01
	(0.19)	(0.16)	(0.18)	(0.16)	(0.17)	(0.17)	(0.22)	(0.10)
4,000–7,500 CNY	0.06	0.05	0.08	0.05	0.07	0.07	0.07	0.05
	(0.24)	(0.22)	(0.27)	(0.22)	(0.25)	(0.25)	(0.26)	(0.23)
7,500–10,000 CNY	0.30	0.28	0.31	0.35	0.29	0.33	0.31	0.31
	(0.46)	(0.45)	(0.46)	(0.48)	(0.46)	(0.47)	(0.46)	(0.46)
10,000–15,000 CNY	0.29	0.32	0.27	0.28	0.30	0.29	0.28	0.30
	(0.45)	(0.47)	(0.45)	(0.45)	(0.46)	(0.45)	(0.45)	(0.46)
15,000–30,000 CNY	0.22	0.23	0.20	0.22	0.22	0.20	0.22	0.21
	(0.41)	(0.42)	(0.40)	(0.41)	(0.41)	(0.40)	(0.41)	(0.41)
Above 30,000 CNY	0.09	0.09	0.11	0.07	0.09	0.08	0.06	0.11
	(0.29)	(0.29)	(0.31)	(0.26)	(0.29)	(0.27)	(0.25)	(0.32)
N	775	127	129	128	130	133	678	743

Note: This table reports descriptive statistics for household characteristics by randomized-experiment group and city. The DCE column pools all DCE respondents across both cities. The 0 to 400 CNY columns correspond to the five randomized-experiment groups and pool respondents across both cities. The Beijing and Shanghai columns pool the DCE and randomized-experiment samples within each city. Standard errors in parentheses.

Table B3: Descriptive Statistics by Group and City: Housing

Variables	DCE	0 CNY	50 CNY	100 CNY	200 CNY	400 CNY	Beijing	Shanghai
House area (m^2)	98.32	98.54	98.15	98.51	98.57	98.41	97.79	99.05
	(33.12)	(35.44)	(34.01)	(34.32)	(28.61)	(31.45)	(30.85)	(34.93)
Equipped with double-layer windows	0.50	0.54	0.53	0.46	0.41	0.45	0.44	0.52
	(0.50)	(0.50)	(0.50)	(0.50)	(0.49)	(0.50)	(0.50)	(0.50)
Equipped with insulation layers	0.53	0.55	0.51	0.51	0.56	0.54	0.64	0.44
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	(0.48)	(0.50)
House built year								
House built before 2000	0.34	0.39	0.31	0.37	0.28	0.30	0.32	0.35
	(0.47)	(0.49)	(0.46)	(0.48)	(0.45)	(0.46)	(0.47)	(0.48)
House built in 2000–2010	0.44	0.41	0.45	0.37	0.49	0.46	0.46	0.40
	(0.50)	(0.49)	(0.50)	(0.48)	(0.50)	(0.50)	(0.50)	(0.49)
House built after 2010	0.22	0.20	0.24	0.26	0.24	0.24	0.22	0.25
	(0.41)	(0.40)	(0.43)	(0.44)	(0.43)	(0.43)	(0.41)	(0.43)
N	775	127	129	128	130	133	702	720

Note: This table reports descriptive statistics for housing characteristics by randomized-experiment group and city. The DCE column pools all DCE respondents across both cities. The 0 to 400 CNY columns correspond to the five randomized-experiment groups and pool respondents across both cities. The Beijing and Shanghai columns pool the DCE and randomized-experiment samples within each city. Standard errors in parentheses.

Table B4: Descriptive Statistics by Group and City: Prosocial Values

Variables	DCE	0 CNY	50 CNY	100 CNY	200 CNY	400 CNY	Beijing	Shanghai
Public spirit (Question 1)								
Very likely	0.27	0.25	0.27	0.30	0.28	0.29	0.29	0.26
	(0.45)	(0.44)	(0.45)	(0.46)	(0.45)	(0.46)	(0.46)	(0.44)
Likely	0.64	0.67	0.63	0.60	0.62	0.61	0.61	0.66
	(0.48)	(0.47)	(0.48)	(0.49)	(0.49)	(0.49)	(0.49)	(0.48)
Not likely	0.08	0.07	0.10	0.09	0.07	0.09	0.09	0.08
	(0.27)	(0.26)	(0.29)	(0.29)	(0.26)	(0.28)	(0.28)	(0.27)
Not at all likely	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.00
	(0.10)	(0.07)	(0.00)	(0.07)	(0.14)	(0.12)	(0.12)	(0.05)
Social norm (Question 2)								
Totally agree	0.14	0.16	0.12	0.14	0.13	0.13	0.12	0.15
	(0.35)	(0.37)	(0.32)	(0.35)	(0.34)	(0.34)	(0.33)	(0.36)
Agree	0.59	0.61	0.60	0.54	0.58	0.57	0.62	0.55
	(0.49)	(0.49)	(0.49)	(0.50)	(0.49)	(0.50)	(0.49)	(0.50)
Disagree	0.25	0.22	0.26	0.29	0.27	0.28	0.24	0.28
	(0.43)	(0.42)	(0.44)	(0.45)	(0.45)	(0.45)	(0.43)	(0.45)
Strongly disagree	0.02	0.01	0.02	0.04	0.01	0.02	0.01	0.02
	(0.14)	(0.07)	(0.14)	(0.19)	(0.12)	(0.16)	(0.12)	(0.16)
Reputational concern (Question 3)								
Not at all likely	0.28	0.31	0.29	0.29	0.23	0.27	0.31	0.24
	(0.45)	(0.46)	(0.46)	(0.45)	(0.42)	(0.44)	(0.46)	(0.43)
Not likely	0.58	0.57	0.53	0.58	0.60	0.58	0.55	0.59
	(0.49)	(0.50)	(0.50)	(0.49)	(0.49)	(0.49)	(0.50)	(0.49)
Likely	0.12	0.11	0.16	0.11	0.15	0.13	0.12	0.14
	(0.33)	(0.32)	(0.37)	(0.31)	(0.36)	(0.34)	(0.33)	(0.35)
Very likely	0.02	0.02	0.02	0.03	0.02	0.02	0.02	0.02
	(0.14)	(0.13)	(0.14)	(0.16)	(0.14)	(0.14)	(0.14)	(0.15)
N	775	127	129	128	130	133	702	720

Note: This table reports descriptive statistics for intrinsic value measures by randomized-experiment group and city. The DCE column pools all DCE respondents across both cities. The 0 to 400 CNY columns correspond to the five randomized-experiment groups and pool respondents across both cities. The Beijing and Shanghai columns pool the DCE and randomized-experiment samples within each city. Standard errors in parentheses.

Table B5: Intrinsic Motivation and Status Quo

	Public spirit (1)	Social norm (2)	Rep. concern (3)
Intrinsic value			
ASC × *public spirit*	-2.893***		
	(0.473)		
ASC × *social norm*		-2.262***	
		(0.394)	
ASC × *rep. concern*			-2.587***
			(0.361)
ASC	0.105	-0.897***	-0.616**
	(0.424)	(0.268)	(0.280)
Other DLC attributes			
$temp_{28}$	-1.131***	-1.193***	-1.187***
	(0.099)	(0.105)	(0.107)
$temp_{29}$	-1.915***	-1.973***	-1.955***
	(0.146)	(0.153)	(0.153)
dur_{120}	-0.588***	-0.623***	-0.600***
	(0.094)	(0.098)	(0.092)
dur_{240}	-0.646***	-0.690***	-0.674***
	(0.113)	(0.114)	(0.111)
company	-0.490***	-0.488***	-0.476***
	(0.100)	(0.099)	(0.094)
gov	-0.541***	-0.566***	-0.563***
	(0.114)	(0.119)	(0.120)
compen	0.006***	0.007***	0.007***
	(0.001)	(0.001)	(0.001)
N	37,200	37,200	37,200
Log likelihood	-2664	-2663	-2659

Note: This table repeats the RPL estimation in Table 5 by adding interaction terms between the ASC and prosocial value measures. The single prosocial value measure terms are absorbed in the RPL model because they do not vary within respondents. We only report the mean estimates. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

C Additional Estimates for Control Variables

Table C1: U-Shaped Relationship in the DCE: with Control Variables

	Dep. var.: DLC alternatives against status quo						
	Full sample	By public spirit		By social norm		By rep. concern	
	Logit	Positive attitudes	Negative attitudes	Positive attitudes	Negative attitudes	Higher concern	Lower concern
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Categorical compensation (base: compensation is 50 CNY)							
compen = 100	-0.204***	-0.205***	0.009	-0.283***	-0.006	-0.297***	0.014
	(0.077)	(0.062)	(0.081)	(0.094)	(0.082)	(0.085)	(0.099)
compen = 200	0.127	0.119	0.213	0.039	0.104	0.072	0.108
	(0.078)	(0.084)	(0.098)	(0.097)	(0.101)	(0.088)	(0.109)
compen = 400	0.412***	0.328***	0.305***	0.303***	0.506***	0.310***	0.441**
	(0.081)	(0.087)	(0.103)	(0.102)	(0.104)	(0.091)	(0.112)
Other DLC attributes							
temp = 28	-0.316***	-0.340***	-0.240	-0.296***	-0.370***	-0.327***	-0.254
	(0.069)	(0.075)	(0.228)	(0.087)	(0.124)	(0.078)	(0.161)
temp = 29	-1.121***	-1.203***	-0.577**	-1.134***	-1.223***	-1.247***	-0.607***
	(0.069)	(0.073)	(0.244)	(0.084)	(0.132)	(0.076)	(0.173)
dur = 120	-0.147**	-0.146**	-0.071	-0.079	-0.285**	-0.122	-0.306*
	(0.068)	(0.074)	(0.233)	(0.085)	(0.125)	(0.077)	(0.165)
dur = 240	-0.204***	-0.221***	-0.068	-0.200**	-0.244*	-0.176**	-0.384**
	(0.068)	(0.073)	(0.232)	(0.084)	(0.126)	(0.076)	(0.165)
company	-0.230***	-0.234***	0.073	-0.172**	-0.290**	-0.246***	-0.313*
	(0.068)	(0.073)	(0.237)	(0.085)	(0.124)	(0.076)	(0.169)
gov	-0.328***	-0.354***	-0.049	-0.358***	-0.227*	-0.323***	-0.341**
	(0.068)	(0.073)	(0.230)	(0.084)	(0.126)	(0.077)	(0.163)
Control variables							
Age (*0.1)	0.026	0.040	-0.200**	0.033	0.020	0.027	-0.006
	(0.024)	(0.026)	(0.081)	(0.030)	(0.046)	(0.027)	(0.060)
Female	-0.025	-0.053	0.120	0.013	-0.121	-0.071	0.090
	(0.058)	(0.062)	(0.202)	(0.072)	(0.107)	(0.064)	(0.151)
Share college grads (*0.1)	-1.213	2.394**	-9.508	1.191	-0.948	0.226	-0.474
	(0.983)	(1.071)	(6.696)	(1.236)	(1.823)	(1.101)	(2.549)
Share <6 or >65 age (*0.1)	-0.006	-0.017	0.046	-0.019	0.130***	-0.009	0.034
	(0.017)	(0.019)	(0.052)	(0.021)	(0.036)	(0.018)	(0.051)
Monthly income >=10k	0.046	0.086	-0.535***	0.059	0.015	0.044	-0.072
	(0.059)	(0.063)	(0.197)	(0.074)	(0.105)	(0.066)	(0.144)
Years of locally living	0.004*	0.011***	-0.046***	0.016***	-0.019***	0.006***	-0.011*
	(0.002)	(0.002)	(0.008)	(0.003)	(0.004)	(0.002)	(0.006)
Housing area (*0.1)	-0.003	0.013	-0.098***	0.004	-0.021	-0.003	0.004
	(0.009)	(0.010)	(0.030)	(0.011)	(0.016)	(0.010)	(0.019)
Double-layer windows	0.097	0.168**	-0.493**	0.053	0.072	0.105	-0.009
	(0.062)	(0.067)	(0.240)	(0.077)	(0.120)	(0.069)	(0.163)
Insulation layer	0.200***	0.249***	-0.320	0.184**	0.147	0.299***	-0.523***
	(0.064)	(0.068)	(0.241)	(0.080)	(0.115)	(0.070)	(0.168)
Built after 2000	0.236***	0.180***	0.075	0.136*	0.245**	0.192***	0.325**
	(0.064)	(0.069)	(0.224)	(0.079)	(0.118)	(0.071)	(0.160)
N	24,800	22,560	2,240	18,112	6,688	21,344	3,456
Log likelihood	-3817	-3359	-332	-2557	-1119	-3104	-635

Note: This table reports the full estimates of Figure 2. The sample size is two thirds of the full DCE sample since the status quo alternatives are dropped. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses. Estimates from specifications without control variables are reported in Appendix Table C2.

Table C2: U-Shaped Relationship in the DCE: without Control Variables

	Dep. var.: DLC alternatives against status quo						
	Full sample	By public spirit		By social norm		By rep. concern	
	Logit	Positive attitudes	Negative attitudes	Positive attitudes	Negative attitudes	Higher concern	Lower concern
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Categorical compensation (base: compensation is 50 CNY)							
$compen = 100$	-0.207***	-0.247***	0.010	-0.289***	-0.003	-0.240***	-0.013
	(0.076)	(0.081)	(0.245)	(0.093)	(0.141)	(0.085)	(0.188)
$compen = 200$	0.130*	0.128	0.147	0.140	0.075	0.127	0.102
	(0.077)	(0.082)	(0.244)	(0.095)	(0.140)	(0.086)	(0.184)
$compen = 400$	0.407***	0.419***	0.395**	0.395***	0.478***	0.404***	0.434**
	(0.080)	(0.086)	(0.146)	(0.100)	(0.141)	(0.090)	(0.188)
Other DLC attributes							
$temp = 28$	-0.302***	-0.319***	-0.201	-0.268***	-0.352***	-0.308***	-0.244
	(0.069)	(0.074)	(0.209)	(0.086)	(0.121)	(0.078)	(0.159)
$temp = 29$	-1.109***	-1.182***	-0.605***	-1.105***	-1.195***	-1.229***	-0.586***
	(0.068)	(0.073)	(0.223)	(0.083)	(0.129)	(0.075)	(0.171)
$dur = 120$	-0.145**	-0.146**	-0.155	-0.072	-0.273**	-0.120	-0.290*
	(0.068)	(0.073)	(0.212)	(0.085)	(0.123)	(0.076)	(0.162)
$dur = 240$	-0.201***	-0.216***	-0.115	-0.193**	-0.237*	-0.172**	-0.373**
	(0.068)	(0.073)	(0.213)	(0.083)	(0.124)	(0.076)	(0.163)
$company$	-0.226***	-0.231***	-0.259	-0.157*	-0.314***	-0.238***	-0.289*
	(0.068)	(0.072)	(0.213)	(0.084)	(0.121)	(0.075)	(0.166)
gov	-0.315***	-0.334***	-0.168	-0.328***	-0.235*	-0.304***	-0.324**
	(0.068)	(0.073)	(0.211)	(0.083)	(0.123)	(0.076)	(0.160)
Control var.	N	N	N	N	N	N	N
N	24,800	22,560	2,240	18,112	6,688	21,344	3,456
Log likelihood	-3839	-3397	-379	-2592	-1147	-3128	-646

Note: This table repeats Appendix Table C1 by excluding control variables. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses.

Table C3: U-Shaped Relationship in the Randomized Experiment

	Dep. var.: willing to participate							
	Full sample		By public spirit		By social norm		By rep. concern	
	Without controls (1)	With controls (2)	Positive attitudes (3)	Negative attitudes (4)	Positive attitudes (5)	Negative attitudes (6)	Higher concern (7)	Lower concern (8)
Base: no compensation								
compen = 50	-0.564***	-0.546***	-0.704***	0.003	-0.713***	0.034	-0.902***	-0.002
	(0.163)	(0.158)	(0.188)	(0.253)	(0.196)	(0.259)	(0.223)	(0.307)
compen = 100	-0.991***	-0.996***	-1.093***	0.008	-1.284***	0.060	-1.154***	0.017
	(0.221)	(0.212)	(0.235)	(0.451)	(0.270)	(0.420)	(0.247)	(0.364)
compen = 200	-0.552**	-0.467**	-0.674***	0.105	-0.670**	0.266	-0.693***	0.201
	(0.218)	(0.211)	(0.235)	(0.450)	(0.274)	(0.409)	(0.248)	(0.353)
compen = 400	0.112	0.123	0.097	0.137	0.084	0.350	0.077	0.350
	(0.236)	(0.213)	(0.261)	(0.469)	(0.300)	(0.416)	(0.262)	(0.330)
Control variables								
Double-layer window		0.062	0.088	0.382	0.072	0.036	-0.008	0.591
		(0.160)	(0.173)	(0.537)	(0.195)	(0.302)	(0.177)	(0.428)
Years of locally living		0.004	0.011*	-0.042**	0.015**	-0.016	0.012**	-0.047***
		(0.005)	(0.006)	(0.020)	(0.007)	(0.010)	(0.006)	(0.016)
House built before 2000		-0.157	-0.184	0.595	-0.011	-0.167	-0.204	0.113
		(0.217)	(0.234)	(0.727)	(0.267)	(0.406)	(0.241)	(0.593)
House built 2000-2010		-0.112	-0.136	-0.055	-0.027	-0.256	-0.155	0.065
		(0.201)	(0.214)	(0.765)	(0.241)	(0.388)	(0.220)	(0.568)
Control variables	N	Y	Y	Y	Y	Y	Y	Y
N	647	647	585	62	463	184	550	97
Log likelihood	-484	-483	-424	-45	-325	-144	-396	-74

Note: This table reports the full estimates of Figure 4. Columns (1) and (2) use the full sample by excluding and including control variables, respectively. Columns (3) to (8) repeat Column (2) by splitting the sample according to attitudes toward intrinsic values. Controls are variables that differ across groups at the 10 percent level or below from the ANOVA test in Table 3. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

Table C4: Wald Tests for DCE

Compensation level	$\Delta\beta$	Wald $\chi^2(1)$	p-value
Panel A: public spirit			
100 CNY	-0.214	4.401	0.036
200 CNY	-0.094	0.530	0.466
400 CNY	0.023	0.029	0.865
Panel B: social norm			
100 CNY	-0.277	4.931	0.026
200 CNY	-0.065	0.215	0.643
400 CNY	-0.203	1.942	0.163
Panel C: rep. concern			
100 CNY	-0.311	5.681	0.017
200 CNY	-0.036	0.066	0.797
400 CNY	-0.131	0.824	0.364

Note: This table reports the Wald tests for Figure 2.

Table C5: Wald Tests for Randomized Experiment

Compensation level	$\Delta\beta$	Wald $\chi^2(1)$	p-value
Panel A: public spirit			
50 CNY	-0.707	5.031	0.025
100 CNY	-1.101	4.687	0.030
200 CNY	-0.779	2.355	0.125
400 CNY	-0.040	0.006	0.941
Panel B: social norm			
50 CNY	-0.747	5.289	0.021
100 CNY	-1.344	7.246	0.007
200 CNY	-0.936	3.615	0.057
400 CNY	-0.266	0.269	0.604
Panel C: rep. concern			
50 CNY	-0.900	5.626	0.018
100 CNY	-1.171	7.086	0.008
200 CNY	-0.894	4.294	0.038
400 CNY	-0.273	0.420	0.517

Note: This table reports the Wald tests for Figure 4.

Table C6: Heterogeneity Analysis: Insulation and Housing Age

	Dep. var.: willing to participate			
	Insulation		Housing age	
	Yes	No	Built after 2000	Built before 2000
	(1)	(2)	(3)	(4)
Base: no compensation				
compen = 50	-0.381*	-0.852***	-0.424**	-1.081***
	(0.201)	(0.247)	(0.215)	(0.227)
compen = 100	-0.821***	-1.647***	-0.922***	-1.504***
	(0.243)	(0.283)	(0.255)	(0.279)
compen = 200	-0.393	-0.894***	-0.417**	-0.896***
	(0.281)	(0.235)	(0.205)	(0.224)
compen = 400	0.142	0.098	0.129	0.114
	(0.287)	(0.263)	(0.232)	(0.247)
Control variables				
Double-layer window	-0.088	-0.011	0.135	-0.064
	(0.236)	(0.265)	(0.198)	(0.280)
Years of locally living	0.000	0.009	0.009	-0.003
	(0.008)	(0.008)	(0.007)	(0.009)
House built before 2000	-0.217	-0.280		
	(0.293)	(0.376)		
House built in 2000–2010	-0.107	-0.181	-0.127	
	(0.243)	(0.375)	(0.202)	
N	346	301	430	217
Log likelihood	-265	-214	-321	-159

Note: This table reports the full heterogeneous logit estimates in Figure 5. Some control variables are used to split the sample and are therefore omitted from the corresponding regressions. The omitted group is no compensation. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

Table C7: Heterogeneity Analysis: Age and Years of Locally living

	Dep. var.: willing to participate			
	Age		Years of residence	
	>=35 yrs old	<35 yrs old	>=20 yrs	<20 yrs
	(1)	(2)	(3)	(4)
Base: no compensation				
compen = 50	-0.423	-1.017***	-0.404	-1.176***
	(0.292)	(0.313)	(0.318)	(0.307)
compen = 100	-0.983***	-1.577***	-1.076***	-1.553***
	(0.328)	(0.303)	(0.297)	(0.302)
compen = 200	-0.463*	-0.957***	-0.577**	-0.924***
	(0.272)	(0.304)	(0.292)	(0.283)
compen = 400	0.118	0.136	0.107	0.131
	(0.251)	(0.278)	(0.244)	(0.269)
Control variables				
Double-layer window	0.460*	-0.141	0.077	0.061
	(0.271)	(0.208)	(0.237)	(0.225)
Years of locally living	-0.007	0.017*	-0.025*	-0.000
	(0.009)	(0.009)	(0.013)	(0.023)
House built before 2000	-0.269	-0.041	-0.234	-0.196
	(0.364)	(0.278)	(0.325)	(0.305)
House built in 2000–2010	0.416	-0.464*	-0.199	-0.153
	(0.343)	(0.256)	(0.313)	(0.271)
N	240	407	331	316
Log likelihood	-176	-294	-237	-238

Note: This table reports the full heterogeneous logit estimates in Figure 5. The omitted group is no compensation. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

Table C8: Heterogeneity Analysis: Gender and Household Education

	Dep. var.: willing to participate			
	Gender		Share of college grads	
	Female	Male	>=0.5	<0.5
	(1)	(2)	(3)	(4)
Base: no compensation				
compen = 50	-0.918***	-1.023***	-1.004***	-0.803**
	(0.287)	(0.358)	(0.303)	(0.337)
compen = 100	-1.217***	-1.343***	-1.296***	-1.104***
	(0.338)	(0.403)	(0.343)	(0.377)
compen = 200	-0.553**	-0.477	-0.597**	-0.452
	(0.283)	(0.347)	(0.288)	(0.332)
compen = 400	0.133	0.112	0.127	0.119
	(0.276)	(0.254)	(0.238)	(0.261)
Control variables				
Double-layer window	0.065	0.044	0.106	0.095
	(0.200)	(0.274)	(0.209)	(0.257)
Years of locally living	0.005	0.003	0.012*	-0.008
	(0.007)	(0.009)	(0.007)	(0.008)
House built before 2000	-0.287	0.113	-0.223	0.025
	(0.276)	(0.360)	(0.279)	(0.356)
House built in 2000–2010	-0.246	0.098	-0.182	0.061
	(0.261)	(0.319)	(0.250)	(0.346)
N	405	242	414	233
Log likelihood	-305	-177	-300	-180

Note: This table reports the full heterogeneous logit estimates in Figure 5. The omitted group is no compensation. $^{*}: p < 0.1$; $^{**}: p < 0.05$; $^{***}: p < 0.01$. Standard errors in parentheses.

D Model Estimation

D.1 Derivation of the Likelihood Function

We start with the logit regression model to introduce our estimation strategy. In conventional logit model, we have the following regression with binary dependent variable y_i:

$$y_i^* = x_i'\beta + \varepsilon_i \tag{12}$$

where continuous y_i^* is the latent variable of real binary y_i. x_i is the dependent variables. β is the vector of coefficients.

We can understand the latent variable y_i^* in the way that $y_i^* = u(y_i = 1) - u(y_i = 0) = x_i'\beta + \varepsilon_i$ where $u(y_i)$ is the utility when i choose y_i. The introduction of latent variable makes the regression with binary dependent variables interpretable. And logit makes the following assumptions:

$$y_i = 1\{y_i^* > 0\} \tag{13}$$

$$\varepsilon_i \sim F(\varepsilon) = \frac{1}{1 + \exp(-\varepsilon)} \tag{14}$$

where F is the Logistic distribution.

The distribution of y_i follows that

$$Pr(y_i = 1|x_i) = Pr(y_i^* > 0|x_i) = Pr(x_i'\beta + \varepsilon_i > 0|x_i) = Pr(\varepsilon_i > -x_i'\beta|x_i) = 1 - F(-x_i'\beta) = F(x_i'\beta) \tag{15}$$

In this sense, given β, the probability that we can observe (y_i, x_i') (which is the definition of likelihood function) i:

$$L(\beta) = \prod_i Pr(y_i = 1|x_i)^{1\{y_i=1\}} Pr(y_i = 0|x_i)^{1\{y_i=0\}} = \prod_i F(x_i'\beta)^{1\{y_i=1\}}(1 - F(x_i'\beta))^{1\{y_i=0\}} \tag{16}$$

where β is $\arg\max_\beta L(\beta)$. We generally take log form to facilitate the maximization of likelihood function.

Next, we leverage this latent dependent variable framework to construct the contribution x^* in Bénabou and Tirole (2006). The optimal choice of x^* is a continuous variable in the model, as interpreted by the level of personal contribution (e.g., the amount of blood donated). However, in our experiment, x is a binary variable to indicate whether the person accepts DLC. In this sense, we regard the x^* in the model as the latent variable of x in our experiment. Recall the solution of x^* in the model is

$$x_i^* = \frac{v_{s,i} + v_{m,i} m_i}{k} + \mu_s \rho(m_i) - \mu_m \chi(m_i) \tag{17}$$

where

$$\rho(m) = \frac{\sigma_s^2 + m\sigma_{sm}}{\sigma_s^2 + 2m\sigma_{sm} + m^2\sigma_m^2}$$

$$\chi(m) = \frac{1 - \rho(m)}{m}$$

This equation is very similar with $y_i^* = x_i'\beta + \varepsilon_i$ but with nonlinear parameters and independent variable of monetary incentive m_i (in other words, all the other parameters than m_i are to be estimated). Given the latent variable framework, to write the likelihood function, we should know $Pr(x = 1|m_i) = Pr(x^* > 0|m_i)$.

Since we have made parametric assumptions on (v_s, v_m) which follows normal distribution:

$$\bar{v} = \begin{pmatrix} \bar{v}_s \\ \bar{v}_m \end{pmatrix}$$

and

$$\Omega = \begin{pmatrix} \sigma_s^2 & \sigma_{sm} \\ \sigma_{ms} & \sigma_m^2 \end{pmatrix}$$

we can directly write $Pr(x^* > 0|m_i)$ by

$$\begin{aligned} Pr(x^* > 0|m_i) &= \mathbb{P}\left(\frac{v_{s,i} + v_{m,i} m_i}{k} + \mu_s \rho(m_i) - \mu_m \chi(m_i) > 0|m_i\right) \\ &\overset{\mathbf{1}}{=} \Phi\left[\frac{(\bar{v}_s + \bar{v}_m m_i)/k + \mu_s \rho(m_i) - \mu_m \chi(m_i)}{\sqrt{(\sigma_s^2 + m_i^2\sigma_m^2 + 2m_i\sigma_{sm})/k^2}}\right] \end{aligned} \tag{18}$$

where equality 1 is derived by knowing that $\frac{v_{s,i}+v_{m,i}m_i}{k} + \mu_s \rho(m_i) - \mu_m \chi(m_i)$ is a normal distribution with mean and standard deviation derived using the property of multivariate normal distribution of (v_s, v_m).

The likelihood function is immediate as equation (11).

D.2 MLE Estimation

Table D1: MLE Estimates of Theoretical Model

	$\bar{v}_s$	$\bar{v}_m$	μ_s	μ_m	σ_s^2
Estimation	0.914	1.703***	0.138	10.838***	2.431**
Std. err.	(0.833)	(0.115)	(1.320)	(2.608)	(1.345)
N			647		
Log likelihood			-345		

Note: This table reports the MLE estimates of model parameters. The parameter k and σ_m^2 are normalized to 1. To avoid under-identification, we further assume $\sigma_{sm} = 0$. $^{*} : p < 0.1$; $^{**} : p < 0.05$; $^{***} : p < 0.01$.

First, although the estimated mean intrinsic-motivation parameter is not significant, this does not imply the absence of intrinsic motivation in the population. The randomized experiment, where nearly 75% of respondents are willing to participate without compensation, already demonstrates the presence of intrinsic motives. The model simply does not require a large average intrinsic-motivation level to fit the data; instead, the observed behavior is primarily driven by heterogeneity across individuals and by the interaction between intrinsic motives and reputational concerns. This is consistent with the theoretical model, in which even modest intrinsic motivation can generate the U-shaped crowding-out pattern.

Second, the signs of the two valuation parameters are consistent with our expectations. The positive estimate of μ_s indicates that a stronger explicit moral self-image makes a positive contribution to personal utility. The positive estimate of μ_m implies that a higher level of explicit greediness reduces utility (since it appears in the utility function as $-\mu_m$), which is consistent with the idea that people dislike being perceived as greedy by their neighbors.

Finally, the significant estimate of the variance parameter is central to the crowding-out mechanism, because a larger variance means that neighbors rely more on inferring your private information. As in the counterfactual analysis, the change in σ's will significantly strengthen or weaken the crowding-out effects.

E Robustness

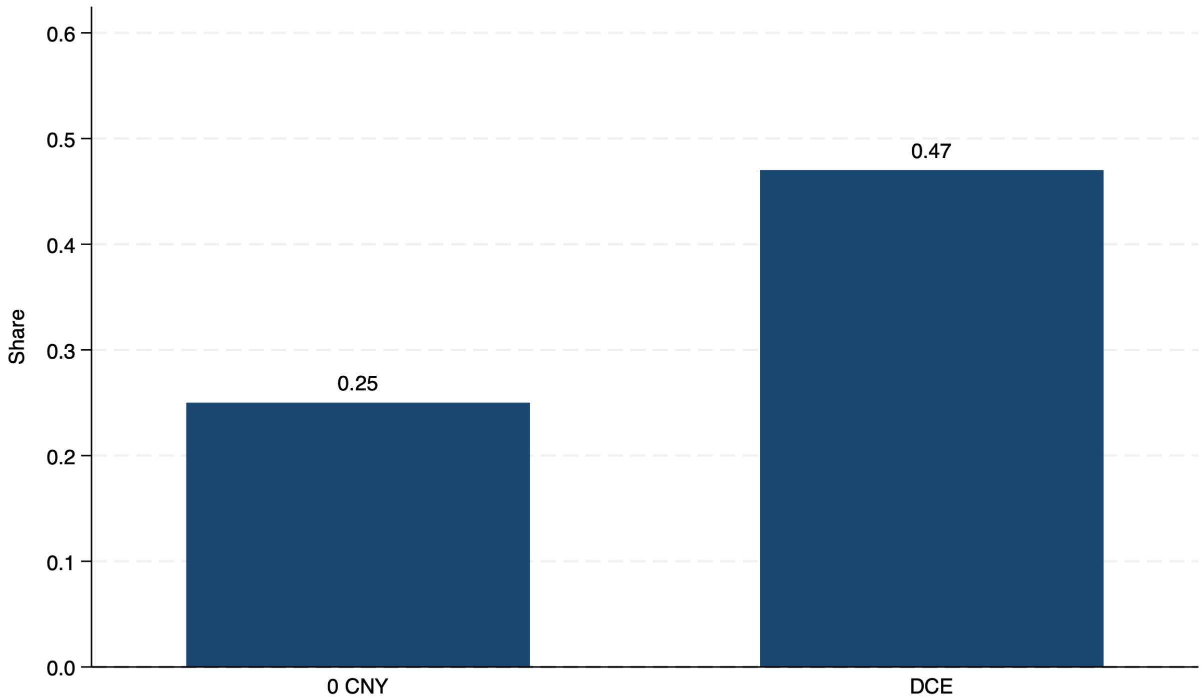

Figure E1: Share of Respondents Stating That Low Compensation Is a Barrier to DLC Acceptance

Note: This figure shows the share of respondents stating that low compensation is a barrier to DLC acceptance after the DCE or randomized experiment. We compare the no-compensation group (0 CNY) in the randomized experiment and the DCE sample.

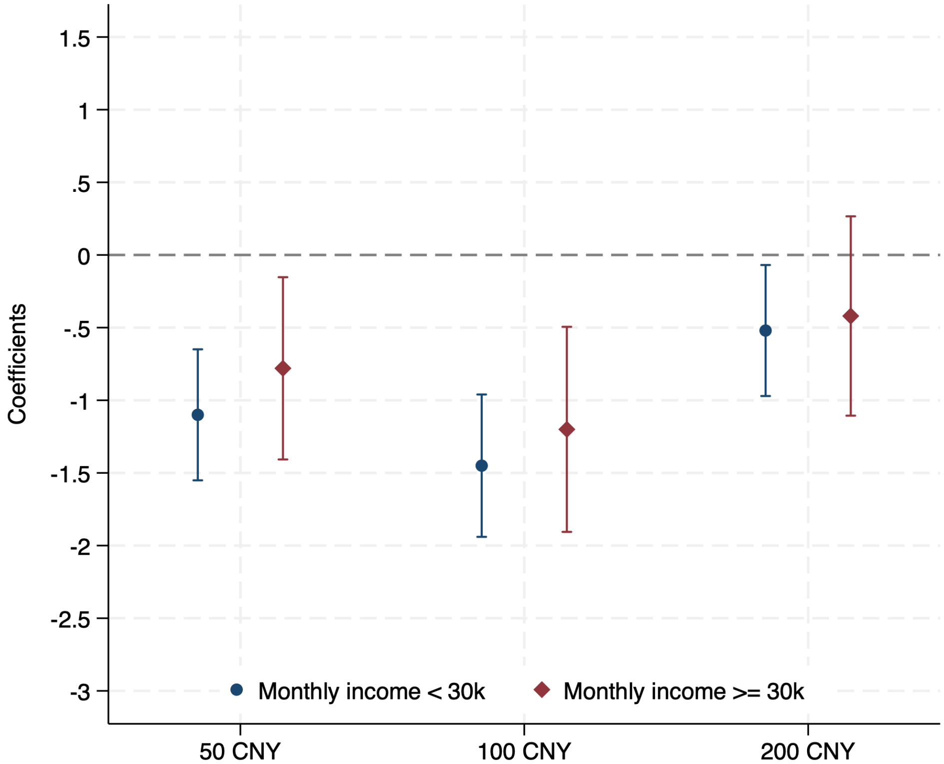

Figure E2: Heterogeneity by Household Income

Note: This figure extends Figure 5 by splitting the sample by household income.